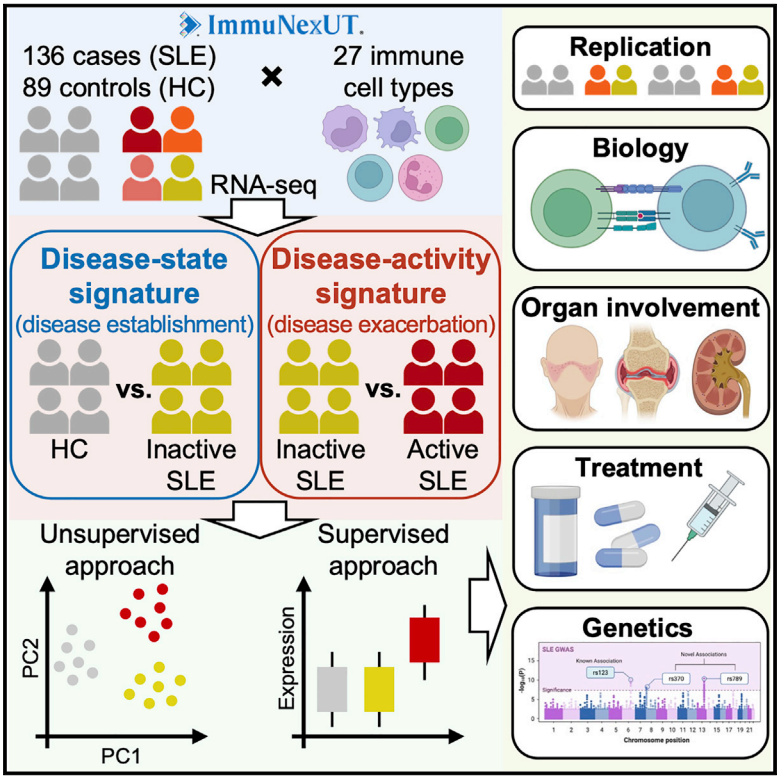
Cell

Distinct transcriptome architectures underlying lupus establishment and exacerbation

Graphical abstract



Highlights

- A large-scale bulk transcriptome study of 27 immune cell types for SLE
- Two distinct gene signatures from case-control and disease-activity comparisons
- Combination of unsupervised and supervised approaches
- Detailed investigation of five major topics, e.g., biology, treatment, and genetics

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In brief

A large-scale transcriptome analysis of 27 immune cell types from systemic lupus erythematous patients and healthy controls reveals two distinct pathological signatures of disease establishment and exacerbation, which could be informative for drug target discovery.

Nakano et al., 2022, Cell *185*, 3375–3389
September 1, 2022 © 2022 Elsevier Inc.
https://doi.org/10.1016/j.cell.2022.07.021

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Distinct transcriptome architectures underlying lupus establishment and exacerbation

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https://doi.org/10.1016/j.cell.2022.07.021

SUMMARY

Systemic lupus erythematosus (SLE) is a complex autoimmune disease involving multiple immune cells. To elucidate SLE pathogenesis, it is essential to understand the dysregulated gene expression pattern linked to various clinical statuses with a high cellular resolution. Here, we conducted a large-scale transcriptome study with 6,386 RNA sequencing data covering 27 immune cell types from 136 SLE and 89 healthy donors. We profiled two distinct cell-type-specific transcriptomic signatures: disease-state and disease-activity signatures, reflecting disease establishment and exacerbation, respectively. We then identified candidate biological processes unique to each signature. This study suggested the clinical value of disease-activity signatures, which were associated with organ involvement and therapeutic responses. However, disease-activity signatures were less enriched around SLE risk variants than disease-state signatures, suggesting that current genetic studies may not well capture clinically vital biology. Together, we identified comprehensive gene signatures of SLE, which will provide essential foundations for future genomic and genetic studies.

INTRODUCTION

Systemic lupus erythematosus (SLE) is a systemic autoimmune disease that involves multiple immune cell types and pathways (Tsokos et al., 2016). SLE has a broad spectrum of clinical manifestations, and the disease course is generally unpredictable (Durcan et al., 2019). This heterogeneous nature has hampered a better understanding of SLE pathogenesis and the

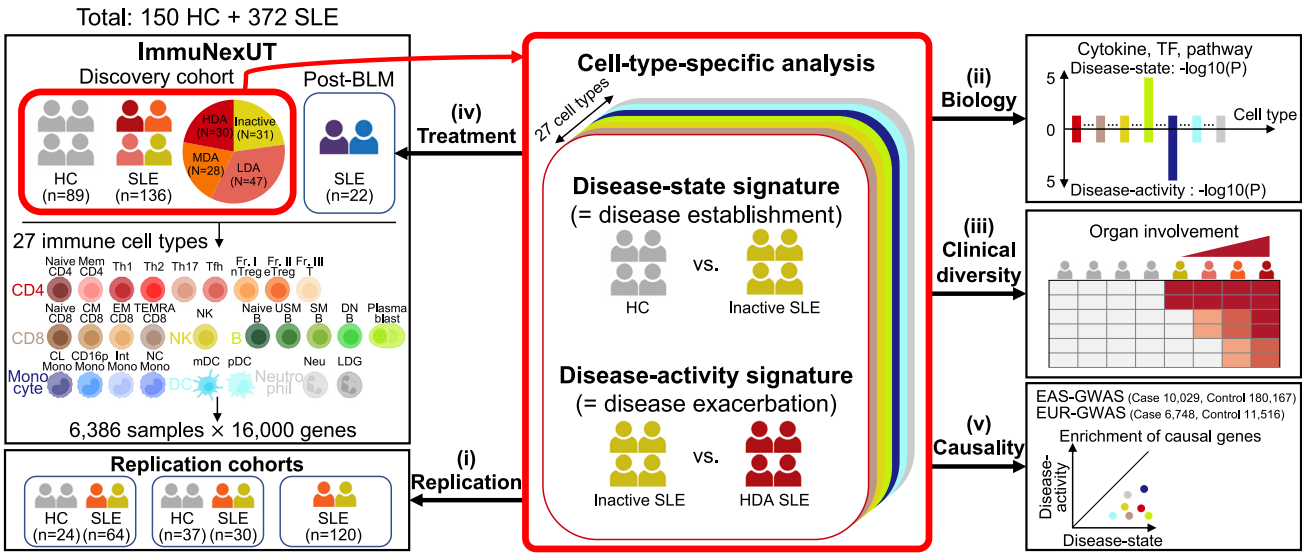


Figure 1. Overview of this study

We profiled 6,386 RNA-seq data of 27 immune cell types from peripheral blood in HC and SLE patients (left). We identified two distinct categories of disease-relevant signatures in a cell-type-specific manner (middle) and then performed extensive downstream analyses (i–v).

HCs, healthy controls; LDA, low disease activity; MDA, moderate disease activity; HDA, high disease activity; BLM, belimumab; TF, transcription factor; EAS, East Asian; EUR, European.

See Table S1A for the description of cell type abbreviations.

development of effective therapeutic agents (Allen et al., 2021; Wallace, 2015). To date, only two biologics have been approved for SLE, belimumab (BLM) and anifrolumab, monoclonal antibodies against B cell-activating factor (BAFF) and type I interferon (IFN) receptor subunit 1, respectively (Morand et al., 2020; Navarra et al., 2011).

To detect biomarkers and therapeutic targets for SLE, several studies on bulk whole-blood transcriptomes have revealed some key gene signatures related to IFN signaling, granulocytes, and plasma cells (Banchereau et al., 2016; Bennett et al., 2003; Panousis et al., 2019). However, these results were biased by the abundance of various immune cell populations in the analyzed samples, which complicates the identification of any cell-type-specific disease-relevant signatures (Lyons et al., 2010). Therefore, recent studies applied single-cell RNA sequencing (scRNA-seq) to peripheral blood mononuclear cells (PBMCs), skin, and kidney samples from SLE patients (Arazi et al., 2019; Der et al., 2019; Nehar-Belaid et al., 2020). While they have successfully identified several cell clusters crucial for lupus pathogenesis, they failed to detect some established cell subpopulations, for example, T helper 1 (Th1), 17 (Th17), and neutrophils (Neu), mainly due to the sparse expression at single-cell level (around 1,000 genes per cell). Moreover, since these scRNA-seq studies of SLE were limited by small sample sizes (21–33 cases), they were not well powered to capture comprehensive transcriptome abnormality related to different clinical manifestations. These limitations could be overcome by a large-scale bulk transcriptome study of multiple established cell subpopulations with higher sensitivity for fine gene signatures linked to precise disease pathophysiology (Combes et al., 2022).

SLE etiology has both genetic and environmental components. Researchers have conducted large-scale genome-wide association studies (GWASs) for SLE (Langefeld et al., 2017; Yin et al., 2021), identifying more than one hundred risk loci. Combined with omics data mostly from healthy individuals, researchers attempted to interpret the genetic etiology and have identified potential causal roles of IFN, Toll-like receptor signaling, and immune complexes (Farh et al., 2015; Westra et al., 2013). However, these studies have not thoroughly investigated the complex interactions between risk variants and the transcriptome dysregulation seen in SLE patients. Such investigations hold the promise to elucidate the complex pathogenesis of SLE.

To address these issues, we conducted a large-scale transcriptome study of 6,386 bulk RNA-seq data including 27 purified immune cell types in peripheral blood that encompassed almost every type of immune cell (Figure 1). We recruited 136 SLE patients with various disease activities and clinical presentations (22 among them were re-evaluated after BLM treatment; STAR Methods) and 89 healthy volunteers in the Immune Cell Gene Expression Atlas from the University of Tokyo (ImmuNexUT) cohort (Ota et al., 2021) (discovery dataset). Using multiple approaches, we investigated cell-type-specific transcriptome dysregulation and classified them into two categories: disease-state and disease-activity signatures. Furthermore, we deployed these signatures to five main topics of downstream analyses: (1) replication, (2) biological interpretation, (3) diverse organ involvement, (4) pre- and post-treatment comparison, and (5) the SLE-GWAS signals (Figure 1). Overall, our large-scale and comprehensive investigation uncovered the molecular basis underlying the clinical heterogeneity of SLE with a fine resolution of cell-type specificity.

RESULTS

Overview of gene expression patterns in the ImmuNexUT cohort

Our dataset included 27 finely sorted immune cell types: CD4 T cells, nine subsets; CD8 T cells, four subsets; natural killer (NK) cells, one subset; B cells, five subsets; monocytes, four subsets; dendritic cells (DCs), two subsets; and neutrophils, two subsets (Figure 1, left; Tables S1A and S1B; STAR Methods). We recruited 248 donors in total. Among them, 136 unique SLE patients and 89 healthy controls (HCs) were included in the discovery dataset; the rest included 22 post-BLM patients (STAR Methods). Compared with previous scRNA-seq studies, a larger sample size with multiple clinical statuses (e.g., disease activity, organ involvement, and treatment profiles) is an advantage of our cohort (Table S1C). At enrollment, 30 patients (22.1%) in the discovery dataset had high disease activity (HDA; SLEDAI-2K $\geq$ 9), while 31 (22.8%) patients were inactive (SLEDAI-2K = 0) (Gladman et al., 2002). Forty-one (30.1%), 27 (19.9%), and 30 (22.1%) patients had mucocutaneous, musculoskeletal, and renal activity, respectively.

To understand the highly complex transcriptomic signatures in our datasets (16,000 genes and 6,386 samples from 248 donors; Figure S1A; Table S1D; STAR Methods), we projected all samples in low dimensional spaces. Principal-component analysis (PCA) and uniform manifold approximation and projection (UMAP) showed that the samples from the same cell type clustered together followed by the same cell lineage (Figures S1B–S1F). We confirmed that batch effects were successfully removed in all cell types (Figures S1G and S1H).

IFN-related gene (IRG) expression is a hallmark signature of SLE (Crow, 2014). To explicitly quantify transcriptome patterns well established for lupus, we utilized 100 IRGs reported in a recent PBMC scRNA-seq study (Nehar-Belaid et al., 2020). We found several cell-type- and disease-specific IRG expression patterns consistent with the original publication (Figure S1I). In addition, about half (n = 54) of IRGs showed the highest expression in SLE neutrophil-lineage cells, which were not evaluated in the previous scRNA-seq studies of SLE. As a result, hierarchical clustering using our own dataset identified new clusters (Figures 2A and S1J): for example, clusters C9 and C11 characterized by the highest expression in SLE neutrophils. Together, our comprehensive transcriptomic data exhibited the expected cell-type specificity and discovered unreported IRG expression patterns.

Disease activity is a major source of variation in the within-SLE transcriptome data

To explore the source of the transcriptomic variations in the discovery dataset, we conducted PCA within each cell type and evaluated the distribution of samples in the PCA space (Table S1E). The top PCs differentiated SLE patients from HC in all cell types, indicating widespread transcriptome perturbations in SLE immune cells (Figures 2B and S2A–S2D). In addition, gene expression profiles within patients showed higher variation than those within HC, reflecting the highly heterogeneous transcriptome patterns in SLE (Figure 2C; STAR Methods).

We further evaluated how the transcriptome variations reflected the heterogeneous clinical statuses. Using PC1–7, the minimum set to associate the transcriptome with the clinical parameters (Figures S2E–S2L; STAR Methods), we quantified the contribution of clinical parameters to the within-SLE variation by weighted variance partitioning analysis (Figure 2D; STAR Methods). Importantly, this analysis revealed that disease activity had the largest contribution to the total variation within SLE in almost all cell types (7.6% on average), around 2.9-fold larger than the treatment contribution.

SLE disease-state and disease-activity signatures

Motivated by the fact that both case-control differences and disease activity substantially contributed to the whole transcriptome architecture, we next deployed a supervised approach to the discovery dataset to identify two transcriptomic signatures for each cell type: (1) disease-state signature genes, defined as differentially expressed genes (DEGs, false discovery rate [FDR] < 0.05) between inactive SLE and HC, reflecting the biology of disease establishment and (2) disease-activity signature genes, defined as DEGs between HDA and inactive SLE, reflecting the biology of disease exacerbation (Figures 1, middle, 3A, S3A, S3D, and S3E). We detected comparable numbers of DEGs between these two comparisons (on average, 2,098 disease-state and 2,114 disease-activity signature genes; Figure S3B; Tables S2A and S2B). We conducted replication analysis using independent cohorts (Andreoletti et al., 2021; Panwar et al., 2021; Takeshima et al., 2022; Figure 1i) and confirmed the robustness of both signatures based on their high replicability (one-sided sign test, Bonferroni-corrected $p < 0.05$; Figures S3G and S3H; Table S2C).

To examine the specificity of these signatures, we first compared them within each cell type. We calculated the Jaccard similarity index in each cell type to quantify the shared genes between signatures (Figure 3A); we considered that a gene was shared when it was included in both signatures with a concordant sign (STAR Methods). Based on the proportion of DEGs and Jaccard index, we found three different transcriptome perturbation patterns (STAR Methods): disease-state dominant pattern, e.g., naive CD4/8 T cells (naive CD4/8), disease-activity dominant pattern, e.g., plasmablasts, and shared pattern, e.g., neutrophil-lineage cells (Figures 3A, 3B, and S3B). When we evaluated the similarity based on the correlation of log fold changes (logFC) for both signatures, we observed consistent patterns (Figure S3C).

We next compared signature genes across different cell types and confirmed cell-type-specific and shared components. Around 20% and 30% of signature genes were detected in only one cell type and lineage, respectively (Figure 3C). To understand the distribution of the shared components across cell types, we calculated the Jaccard index (Figure 3D) and correlation (Figure S3F). Globally, we detected higher similarity within the same cell lineage than in different lineages in both signatures. However, we found a clear discrepancy between the signatures; high similarity among Th1 and cytotoxic lymphocytes (NK and memory CD8-lineage cells) (Uzhachenko and Shanker, 2019) was observed only in activity signatures (blue square in

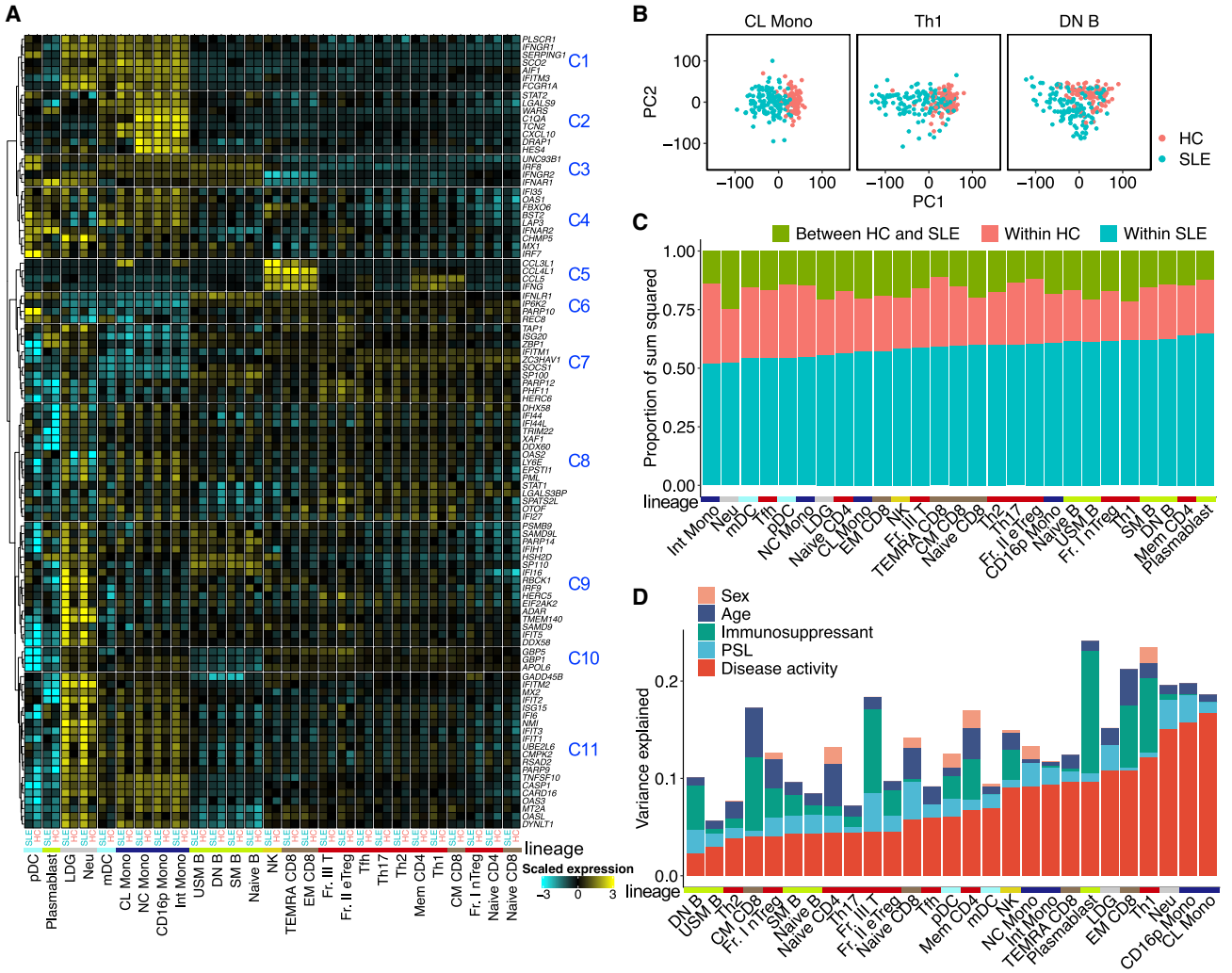


Figure 2. Overview of gene expression patterns in the ImmuNexUT cohort

(A) A heatmap showing the mean expression of 100 IRGs across all cell types and diseases. Expression levels are scaled for each gene. Genes and cell types are hierarchically clustered based on the IRG expression levels of our dataset.

(B) PCA plots of HC and SLE RNA-seq data in representative cell types (see also Figure S2A).

(C) A bar plot showing the proportion of sum-squared deviations within HC, SLE, and between HC and SLE data in each cell type. Cell types are arranged based on the sum-squared deviations within SLE.

(D) A bar plot showing the proportion of variance explained by the clinical parameters within-SLE data in each cell type. Cell types are arranged based on the variance explained by disease activity. In (A), (C), and (D), column annotation colors indicate cell lineages. We analyzed the discovery dataset (n = 225) in this figure. PSL, prednisolone.

Figures 3D and S3F). These gaps indicated the presence of gene expression patterns specific to HDA patients in these cell types. Thus, our dataset captured distinct transcriptome perturbations in the disease establishment and exacerbation phases in a cell-type-specific manner.

Cell-type-specific biology in disease establishment and exacerbation

To understand the SLE biology in disease establishment and exacerbation, we next sought to interpret disease-state and activity signature genes in each cell type using multiple external databases (Figure 1ii).

First, we focused on 136 genes encoding cytokines (Table S3A), the key regulators of immune responses and potential drug targets in autoimmune diseases (Dean et al., 2000). Among 51 genes upregulated in at least one signature, we identified 21 and 17 upregulated specifically in one cell lineage of disease-state and activity signatures, respectively (Figure 4A, top). Representative examples of activity signatures included *IL12A/B* (switched memory B cells [SM B]), *IL1B/CCL2/CCL8* (monocyte lineage), and *IL18/TNFSF15* (neutrophil lineage). Among them, *IL21* and *CXCL13*, critical genes to support B-cell antibody production (Rao, 2018), are especially intriguing. Although previous studies reported T follicular helper (Tfh) cells as the major source

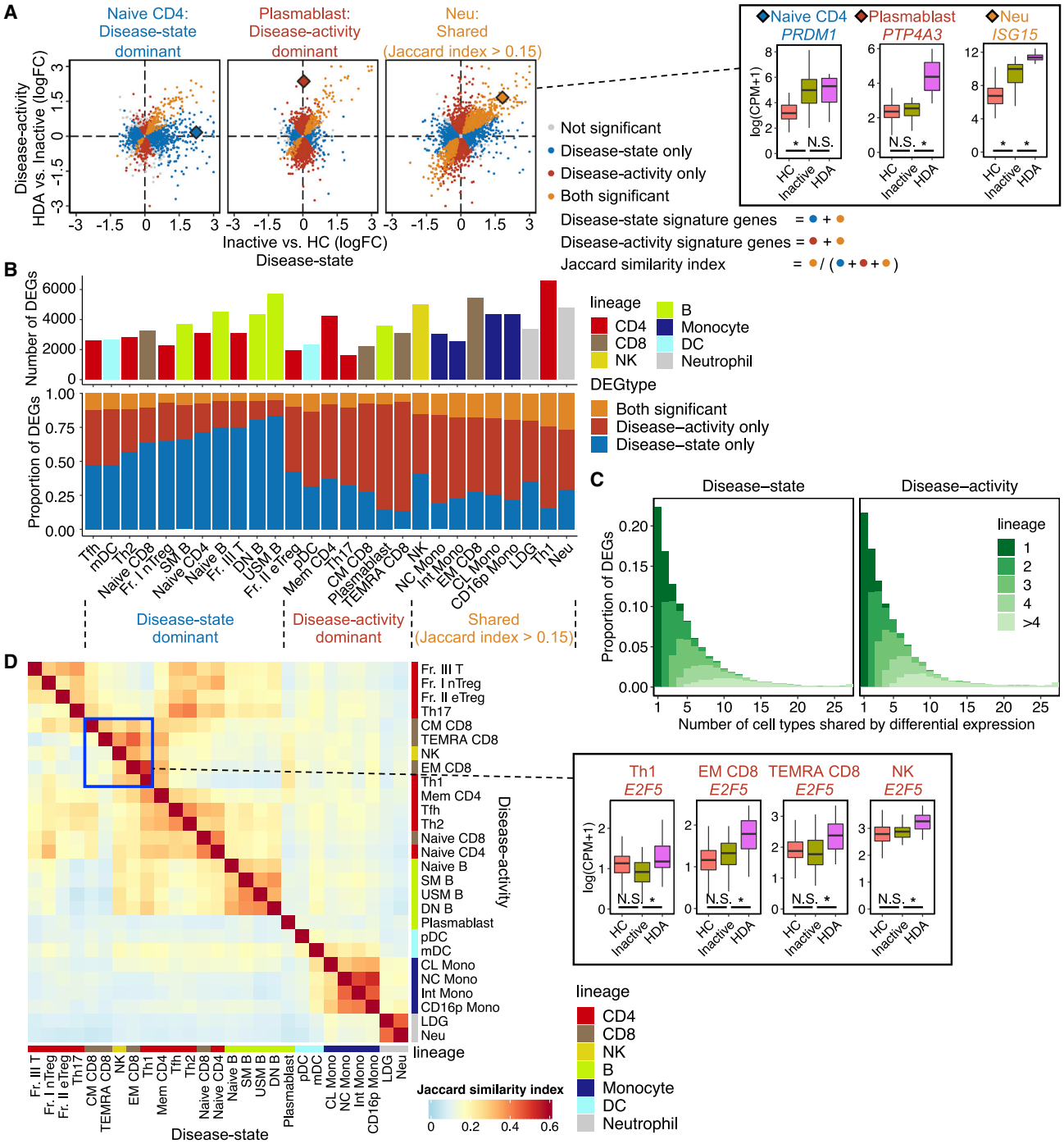


Figure 3. SLE disease-state and disease-activity signatures

(A) (Left) Scatter plots comparing the logFC of disease-state and disease-activity signature genes in representative cell types (see also Figure S3A). Colors indicate the significance of each signature. (Right) Box plots showing the expression of representative disease-state, activity, and both significant signature genes.

(B) Bar plots showing (top) the number of the union of disease-state/-activity signature genes and (bottom) the proportion of DEG types in each cell type. Cell types are separated into three groups.

(C) Histograms showing the proportion of the number of cell types sharing DEG for both signatures. Colors indicate the number of shared cell lineages.

(D) (Left) A heatmap showing the Jaccard similarity indexes across all cell types in both signatures. The order of cell types in row and column is the same and based on the hierarchical clustering using the Jaccard indexes of disease-state signatures. (Right) Box plots showing the expression of a representative disease-activity signature gene shared by Th1, NK, and memory CD8-lineage cells. *DEG (FDR < 0.05); N.S., not significant. We analyzed the discovery dataset (n = 225) in this figure.

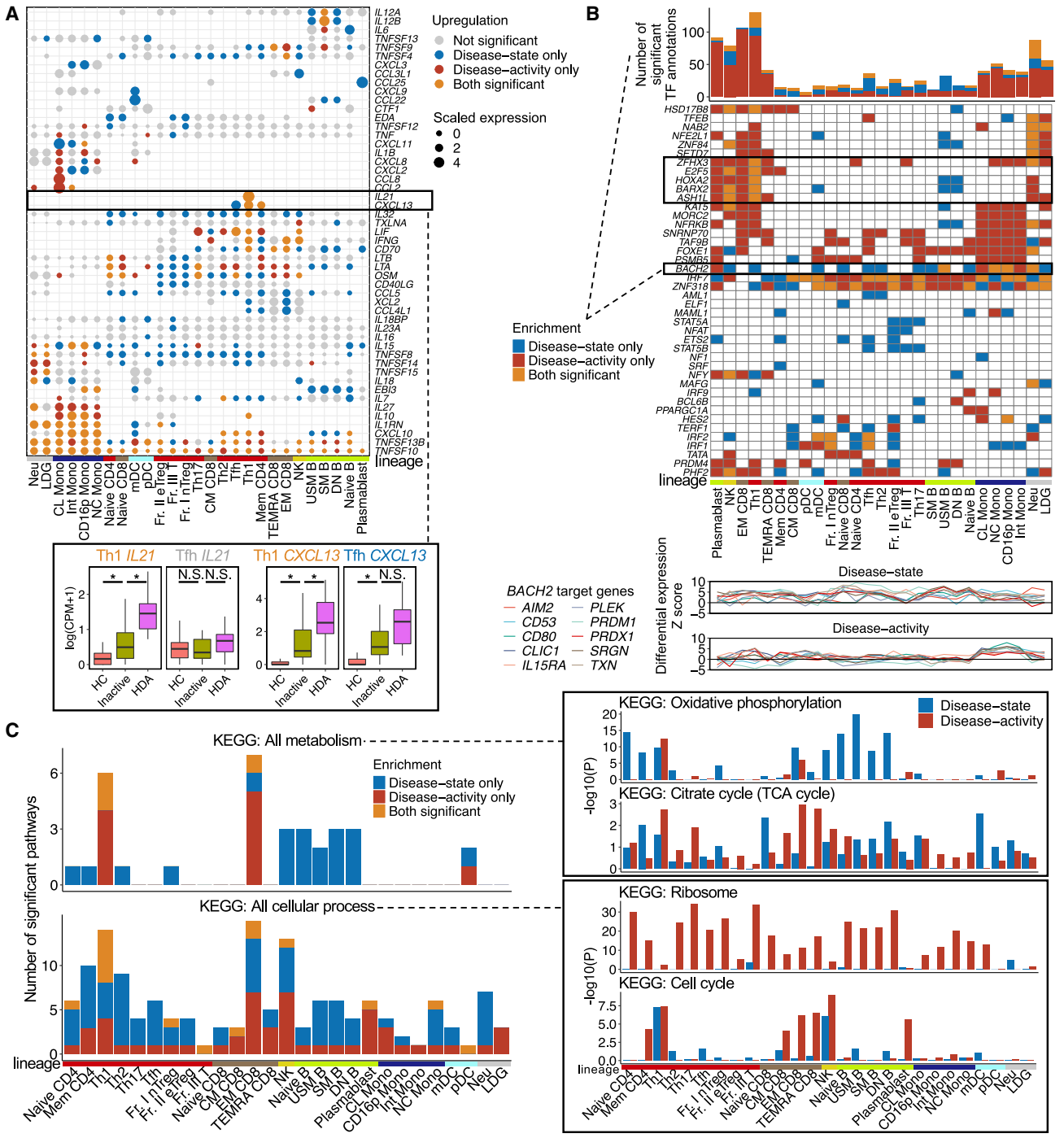


Figure 4. Cell-type-specific biology in disease establishment and exacerbation

(A) (Top) Upregulated cytokines as disease-state and/or disease-activity signature genes for each cell type. Genes and cell types are hierarchically clustered based on differential expression *Z* scores of activity signatures. (Bottom) Boxplots showing the expression of representative cytokines. *DEG (FDR < 0.05); N.S., not significant.

(B) (Top) A bar plot showing the number of significant TF annotation enrichments for each signature. (Middle) A heatmap showing TF enrichment for each signature. TFs and cell types are hierarchically clustered based on $-\log_{10}$(enrichment p) of activity signatures. Only the top three TFs with the strongest enrichments in each signature are shown, excluding redundant annotations. (Bottom) Line graphs showing the differential expression *Z* scores of 10 representative *BACH2* target genes in each cell type for both signatures.

(C) Bar plots showing (left) the number of significant enrichments of metabolism- and cellular process-related pathways and (right) the enrichment of representative pathways for each signature. p, p values in one-sided Fisher's exact test. We analyzed the discovery dataset (n = 225) in this figure.

of *IL21* and *CXCL13* (Crotty, 2019), Th1 showed a more dynamic increase than Tfh, especially in the activity signatures (Figure 4A, bottom). Cell–cell communication analysis prioritized the key ligand-receptor networks that could induce the disease-state and activity signatures in each cell type (Figure S4A and S4B).

Next, we inferred the activities of transcription factors (TFs), essential regulators of immune function, based on the expression of TF-downstream genes (STAR Methods). Among 61,182 total tests (1,133 TF annotations × 27 cell types × two signatures), we observed 1,228 significant enrichments for 299 annotations (FDR < 0.05; one-sided Fisher's exact test; Table S3B). Disease-activity signatures showed more enrichments (862 enrichments [70.2%]; Figure 4B, top). We here highlight two TFs as the candidate key regulators in SLE exacerbation (Figure 4B, middle). First, cell cycle regulators including E2F families showed strong enrichment in activity signatures of Th1, NK, memory CD8-lineage cells, and plasmablasts. Upregulated cell cycle regulation might be driving the high similarities among these cell types in disease-activity signatures (Figure 3D, right). Second, *BACH2* showed significant enrichment in disease-activity signatures for monocyte- and neutrophil-lineage cells. Intriguingly, *BACH2* also showed strong enrichment in lymphocytes, consistent with previous studies (Kometani et al., 2013; Sidwell et al., 2020) but primarily in disease-state signatures. These results demonstrated that an identical gene regulatory machinery can exert a pathogenic effect in different cell types depending on the disease phases (Figure 4B, bottom).

Finally, we performed pathway enrichment analyses to examine multiple biological processes underlying lupus pathogenesis (STAR Methods). Among 32,292 total tests for 598 pathways, we observed 735 and 315 significant enrichments for disease state and disease activity, respectively (FDR < 0.05; one-sided Fisher's exact test; Figure S4C; Tables S3C and S3D). We confirmed the enrichments of established hallmark pathways for SLE (Figure S4D). In addition, we found different enrichment patterns between the signatures in metabolism- and cellular process-related pathways (Figure 4C, left and S4E; STAR Methods). Oxidative phosphorylation signaling was enriched mostly in disease-state signatures (e.g., B-lineage cells), whereas citrate cycle signaling showed enrichment in activity signatures (e.g., memory CD8-lineage cells; Figure 4C, right). Ribosome and cell cycle pathways (e.g., Th1, NK, and memory CD8-lineage cells) were enriched predominantly in disease-activity signatures. Our cell-type-specific analysis successfully clarified the cell-type origin of these pathways and highlighted the importance of previously underappreciated molecular pathways such as immunometabolism, describing disease establishment and exacerbation phases separately.

Cell-type-specific contribution to organ involvement in SLE

To resolve the complex relationships between transcriptome dysregulation and clinical heterogeneity in SLE, we leveraged a PC-based unsupervised approach (Figure 1iii). In the hierarchical clustering of 225 unique individuals using all 189 PCs (=7 PCs × 27 cell types), HC was clearly separated from patients; in addition, HDA patients with multiple organ complications were clustered together (Figures 5A and S5A).

Compared with the approach using thousands of signature genes, the PC-based approach is better at representing whole transcriptome architecture with a small number of parameters. To understand the biological significance of each PC, we defined two categories of PCs as in the discussion of signature genes: (1) disease-state PCs, separating inactive SLE and HC and (2) disease-activity PCs, separating HDA and inactive SLE (FDR < 0.05; Figure 5B; Table S4A; STAR Methods). Among 189 PCs, we identified 37 disease-state PCs and 25 disease-activity PCs; among them, nine PCs were classified into both (Figure S5B). When we projected the data from external cohorts onto our PCA space, PC scores maintained the original contrasts, confirming the good replicability of both PC signatures (one-sided sign test, $p < 0.05$; Figure S5C; Table S4B). This PC-based approach successfully captured the continuous nature of SLE biology; most disease-activity PCs showed a gradual increase in the association signals along with the extent of disease activity (Figures 5B and S5B).

To evaluate cell-type-specific contributions to organ involvement, we assessed the variance proportion of cell-type-specific PCs explained by clinical parameters (weighted variance partitioning analysis; STAR Methods). Overall disease activity, a composite measure reflecting the status of all organs, significantly contributed to the within-SLE transcriptome variation especially in the 13 cell types including Th1, plasmablasts, and monocyte- and neutrophil-lineage cells (Bonferroni-corrected jackknife resampling p [p_{jk}] < 0.05; Figures 2D and 5C, left; STAR Methods). We then decomposed the overall activity into seven organ/domain categories (Isenberg et al., 2005): constitutional, mucocutaneous, musculoskeletal, renal, extrarenal severe, hematological, and serological activities. Interestingly, each organ/domain showed cell-type-specific patterns (Figures 5C, right and S5D–S5H; Tables S4C and S4D; STAR Methods). While Th1 showed the highest explained variance for mucocutaneous activity, the contribution of monocyte-lineage cells was predominant for musculoskeletal activity (Figure 5C, right). Furthermore, neutrophil-lineage cells exhibited the largest contribution to renal involvement, followed by monocyte-lineage cells, Th1, and plasmablasts.

To supplement the variance partitioning analysis based on multiple PCs, we also evaluated the specific relationship of each disease-activity PC with organ involvement (Figure 5D; Table S4A). This analysis highlighted the contribution of (1) Neu and nonclassical monocyte (NC Mono) to renal and (2) naive CD4 and double-negative B cells (DN B) to musculoskeletal involvement (linear regression test; FDR < 0.05). Together, our results confirmed the critical roles of granulocytes and macrophages for the development of lupus nephritis (Gupta and Kaplan, 2016; Kuriakose et al., 2019) and suggested other potential cell-type-specific contribution to organ involvement, which may be informative in unraveling SLE clinical heterogeneity.

Cell-type-specific activity signatures linked to treatment responses

BLM is a monoclonal antibody that inhibits BAFF, a vital factor for B-cell survival and differentiation (Vincent et al., 2014). We investigated the effect of BLM on the transcriptome in each

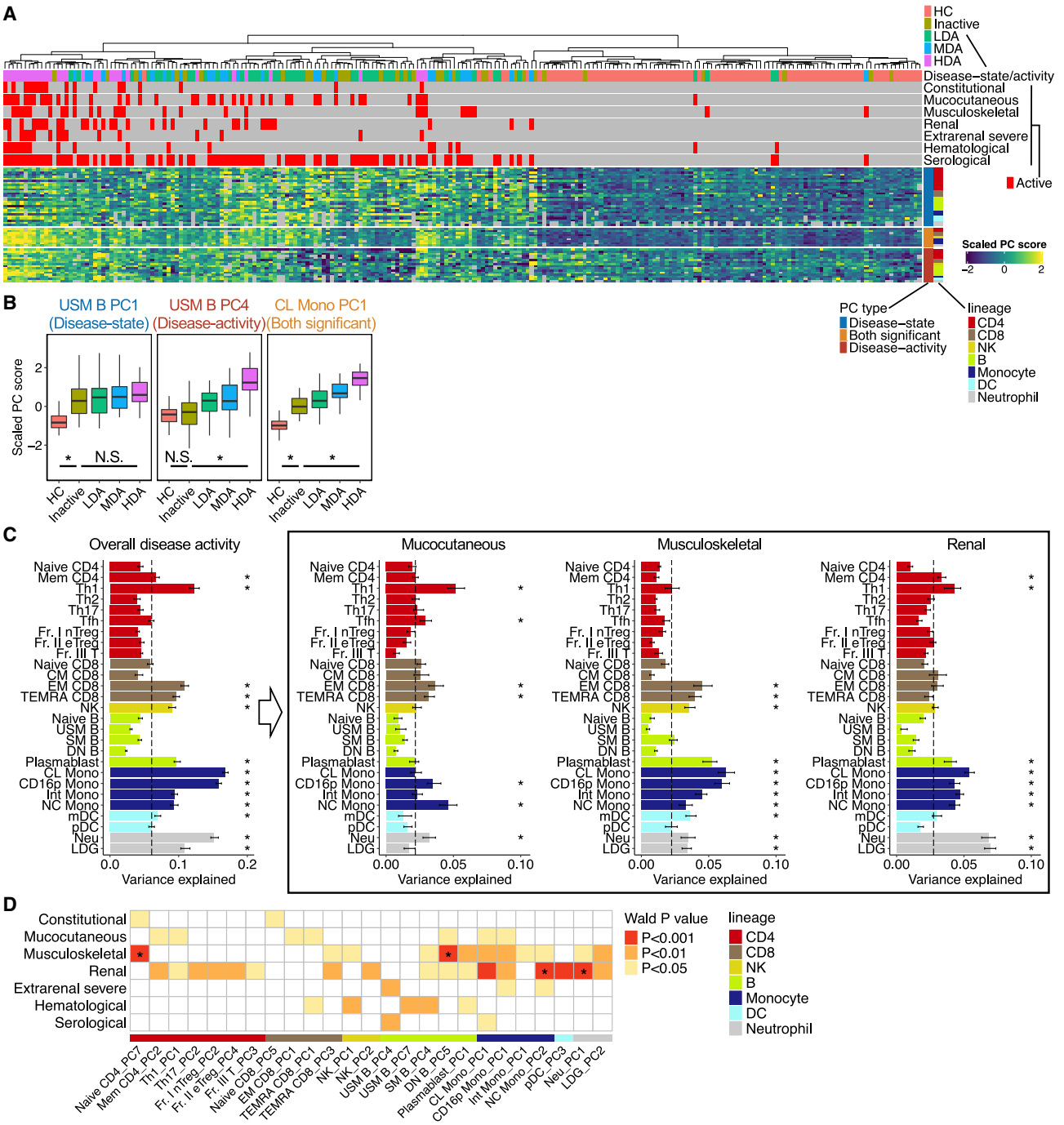


Figure 5. Cell-type-specific contribution to organ involvement in SLE

(A) Hierarchical clustering of 225 unique individuals based on all PC1–7 scores of 27 cell types. Top annotations indicate the disease status and organ/domain activities in each individual. Right annotations indicate the type and cell lineage of each PC. Here, only disease-state and/or activity PCs are shown in the heatmap (see also Figure S5A).

(B) Box plots showing the scaled PC scores in representative disease-state, -activity, and both significant PCs. *FDR < 0.05 in linear regression test; N.S., not significant.

(C) Bar plots showing the proportion of variance explained by (left) the overall disease activity and (right) representative organ activities within-SLE data in each cell type (see also Figure S5E). Error bars and dashed vertical lines indicate 95% confidence intervals from jackknife resampling and the median values across 27 cell types, respectively. *Bonferroni-adjusted $p_{jk} < 0.05$.

(D) A heatmap showing the association of disease-activity PCs and organ activities in SLE. p, nominal p values; *FDR < 0.05 in linear regression test. We analyzed the discovery dataset (n = 225) in this figure.

cell type (Figure 1iv). Our cohort has longitudinal data before and 6 months after BLM induction on 22 individuals; we refer to them as pre- and post-BLM. Importantly, none of the post-BLM samples were included in the discovery dataset; therefore, the comparison between pre- and post-BLM is independent of the disease-activity signatures calculated in the discovery dataset. We observed DEGs between pre- and post-BLM (BLM-DEGs) predominantly in B-lineage cells, confirming the cell-type-specific effects of BLM (Figure 6A). When we categorized the patients into good and poor responders to BLM (n = 9 and 13, respectively; STAR Methods), more DEGs were observed in good than in poor responders (Figures 6B and S6C; Tables S5A and S5B). The IFNγ-related, nuclear factor-kappa B (NF-κB)-related, and glycolysis-related pathways were enriched in B-cell DEGs of good responders (FDR < 0.05, one-sided Fisher's exact test; Figure 6C; Table S5C), consistent with the downstream signaling of BAFF receptors in B cells (McAllister and Jellusova, 2021; Sun, 2017).

We next asked whether BLM effects on transcriptomes counteract disease-activity signatures. We first calculated the Jaccard similarity index to quantify the shared genes between BLM-DEGs and disease-activity signatures; we considered a gene is shared when the BLM effect had the opposite sign to the activity signature to reflect therapeutic responses (STAR Methods). Jaccard indexes in good responders were around 3.4-fold higher than those in poor responders in B-lineage cells (Figure 6B). The analyses based on logFC also showed similar trends (Figures S6A and S6B). By projecting post-BLM data onto the PCA space of the discovery cohort (STAR Methods), we evaluated the change in disease-activity PC scores between pre- and post-BLM. Consistent with the DEG analysis, we found a significant decrease in unswitched memory B cells (USM B) PC4 scores only for good responders (linear mixed regression test, FDR = 9.9×10^{-3}; Figure 6D). Together, these results provided robust evidence supporting the association between disease-activity signatures and the therapeutic response to BLM.

Next, we assessed the effect of other therapeutic agents such as mycophenolate mofetil (MMF) on disease-activity signatures (Figures 6E and S6E; Table S5D). Importantly, we treated disease activity as a potential confounder in this analysis (STAR Methods); hence, the MMF effects are not biased by disease activity. DEGs between patients with and without MMF (MMF-DEGs) were primarily observed in plasmablasts, followed by Th1 and central memory CD8 T cells (CM CD8) (Figure 6E); the same cell types were nominated by variance partitioning analysis (Figure S6F; Table S4C). MMF-DEGs were shared with the activity signature genes with 15.0%–21.7% of the Jaccard index (Figure 6E). Pathway analysis recapitulated this shared component: both genes showed enrichment for oxidative phosphorylation and E2F-related cell cycle pathways (Figure S6D; Table S5E). We also found a significant decrease in disease-activity PC scores of plasmablasts (PC1) in patients taking MMF, adjusted for disease activity (linear regression test, FDR = 6.3×10^{-3}; Figure 6F; STAR Methods). These results were consistent with previous reports that MMF suppressed plasma cell differentiation (Jonsson and Carlsten, 2003).

Risk variants for SLE are enriched around disease-state signatures, not disease-activity signatures

To estimate the causal roles of these signatures for the risk of disease onset, we integrated our transcriptome data with the results of GWASs for SLE (SLE-GWASs; Figure 1v).

First, we analyzed the genome-wide distribution of all risk variants irrespective of their effect sizes and tested their enrichment around the signature genes using stratified linkage disequilibrium score regression (S-LDSC) (Finucane et al., 2015, 2018; STAR Methods). To be in line with previous studies of S-LDSC, we additionally took a conventional approach and analyzed the enrichment of specifically expressed genes (SEGs) in each cell type derived from HC (HC-SEG; STAR Methods). Since ancestry-specific GWAS results are required in S-LDSC, we used two large-scale SLE-GWASs conducted in East Asian (EAS) (Yin et al., 2021) and European (EUR) populations (Langefeld et al., 2017). We confirmed that the S-LDSC results for EAS- and EUR-GWASs were globally similar as reported for other traits (Amariuta et al., 2020) (r in enrichment estimate = 0.69; $p = 1.5 \times 10^{-12}$; Figure S7A; Table S6A); hence, we combined them using a fixed-effect meta-analysis. For HC-SEG, there was no significant enrichment at Bonferroni-corrected p < 0.05 (minimum nominal p = 0.028 for SM B; Figure 7A). Strikingly, compared with HC-SEG, we found much stronger enrichments for disease-state signatures in all cell types: nine significant enrichments. However, for disease-activity signatures, we found no significant enrichment. Consistently, the enrichments were substantially weaker in the activity signatures than in the disease-state signatures (paired Wilcoxon test, $p = 5.5 \times 10^{-6}$).

We next analyzed prioritized causal genes implicated in SLE-GWASs (Table S6B; STAR Methods) and tested their enrichment in both signatures of each cell type (Figures 7B and S7B; Table S6C). As in the S-LDSC results, the enrichments of the GWAS-prioritized genes were predominantly observed in disease-state rather than in activity signatures; significant enrichments were found in three and zero cell types, respectively (Bonferroni-corrected p < 0.05). Again, the enrichments were weaker in activity signatures than in disease-state signatures (paired Wilcoxon test, $p = 2.6 \times 10^{-5}$). These results confirmed that the risk variants locate on average around disease-state and not around disease-activity signature genes.

Considering that most of the risk variants are within the gene regulatory regions (Farh et al., 2015), we hypothesized that risk variants possess gene regulatory effects on disease-state genes (i.e., expression quantitative trait loci [eQTL] effects; Figure 7C). To test this hypothesis, we next evaluated the potential eQTL effects of the risk variants on disease-state and activity signature genes. Specifically, we focused on genes affected by the risk variants' eQTL effects (eGenes) and asked how the risk alleles' directional effects on eGenes are consistent with our gene expression signatures; an eGene with consistent direction is called a "coherent gene" (Marigorta et al., 2017) (Figure 7D; for this analysis, we utilized our recent colocalization test results between SLE-GWAS and eQTL variants [416 donors]) (Ota et al., 2021). Intriguingly, the coherent genes were enriched in disease-state signatures but not in activity signatures (Figure 7E; Table S6D): 67% of eGene-cell

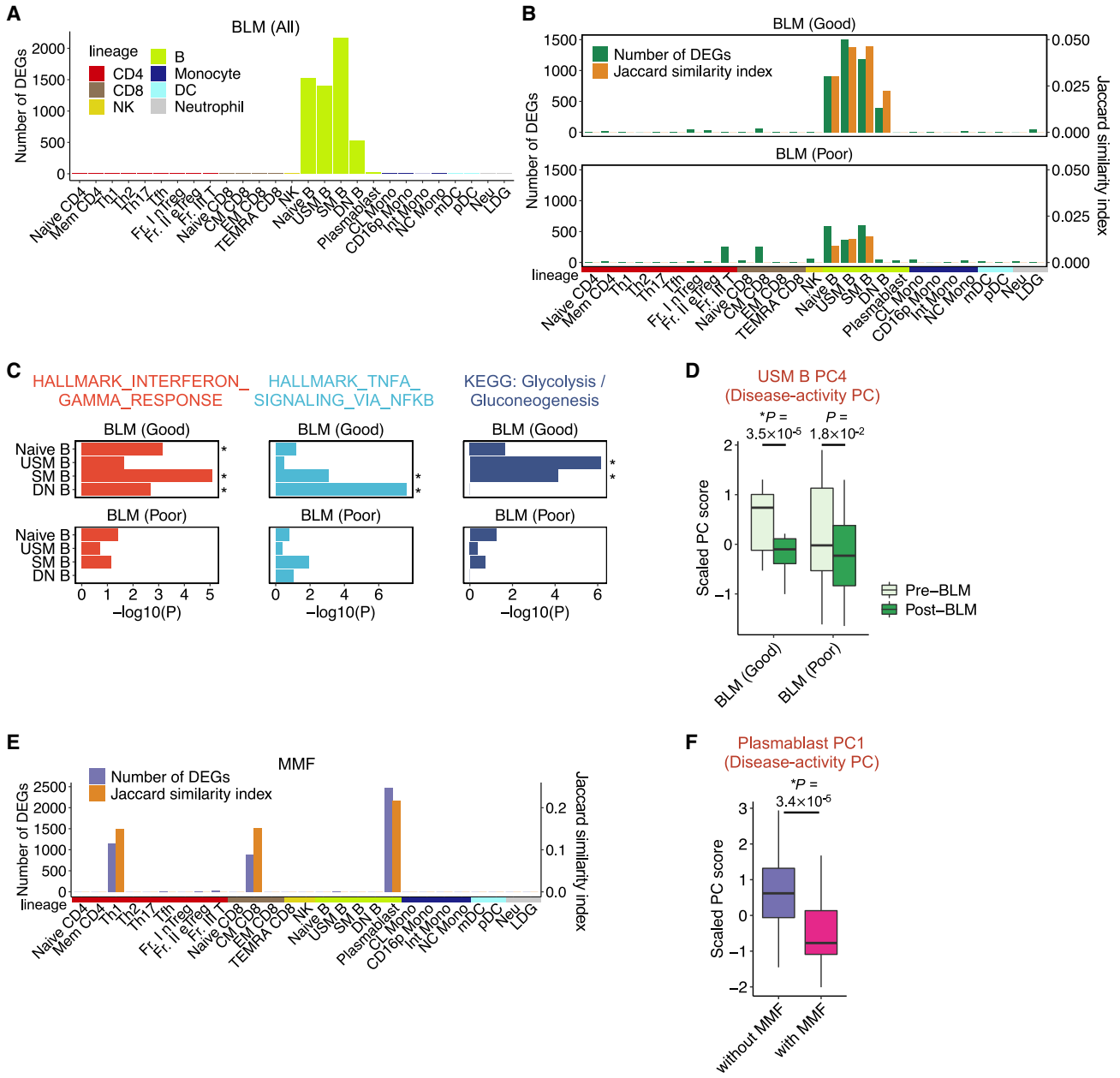


Figure 6. Cell-type-specific activity signatures linked to treatment responses

(A) A bar plot showing the numbers of DEGs in each cell type between all post- versus pre-BLM patients (n = 22 paired samples).

(B) Bar plots showing the numbers of BLM-DEGs and Jaccard similarity indexes between BLM-DEGs and disease-activity signature genes in each cell type, separated into good (n = 9) and poor responders (n = 13).

(C) Bar plots showing the enrichment of representative pathways for the BLM-DEGs in B-lineage cells. p, nominal p values; *FDR < 0.05 in one-sided Fisher's exact test.

(D) A box plot showing the USM B PC4 scores from pre- and post-BLM, separated into good and poor responders. p, nominal p values; *FDR < 0.05 in linear mixed regression test.

(E) Bar plots showing the numbers of DEGs between patients with (n = 31) and without (n = 105) MMF (MMF-DEGs) and Jaccard indexes between MMF-DEGs and disease-activity signature genes in each cell type.

(F) A box plot showing the plasmablasts PC1 scores from patients with or without MMF. p, nominal p values; *FDR < 0.05 in linear regression test.

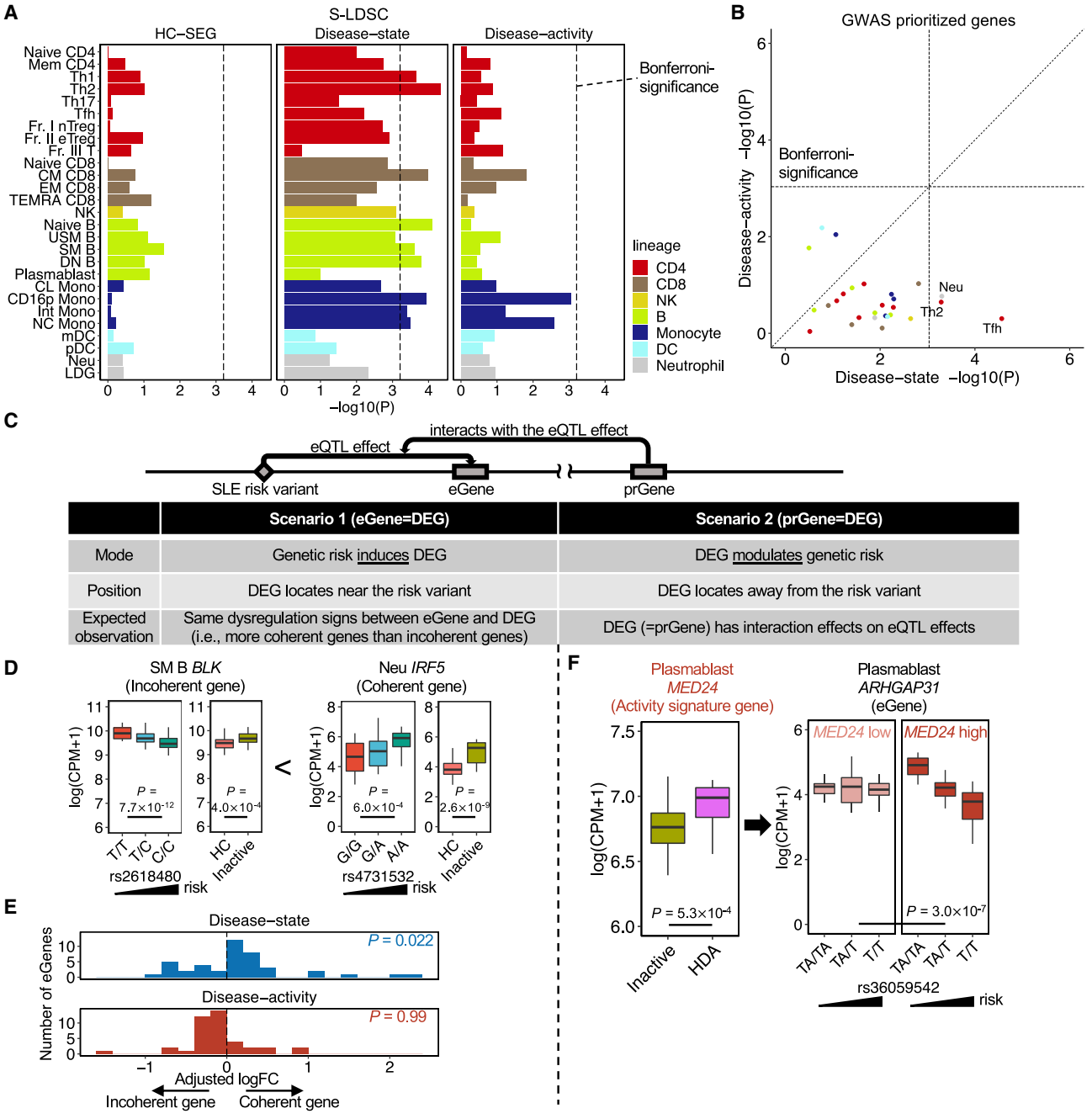


Figure 7. Risks variants for SLE are enriched around disease-state signatures, not disease-activity signatures

(A) Bar plots showing the enrichment of SLE risk variants around HC-SEG, disease-state, and disease-activity signatures for each cell type. p, enrichment p values in S-LDSC. Vertical dashed lines represent Bonferroni significance.

(B) A scatter plot comparing the enrichment of SLE-GWAS prioritized genes in disease-state and disease-activity signatures for each cell type. Only the cell types that pass Bonferroni significance (dashed lines) are annotated. p, p values in one-sided Fisher's exact test.

(C) Hypothesis of the association between the risk variants and DEGs in SLE via eQTL effect.

(D) Box plots showing the expression patterns in representative coherent and incoherent genes. p, p values in linear regression (left) and differential expression test (right).

(E) Histogram of adjusted logFC in coherent and incoherent genes for disease-state and disease-activity signatures. p, p values in one-sided sign test.

(F) Box plots showing (left) the differential expression of *MED24*, an activity signature gene in plasmablasts and (right) the influence of *MED24* on the eQTL effect of an SLE risk variant rs36059542 on *ARHGAP31*. p, p values in differential expression (left) and ANOVA test (right). We analyzed the discovery dataset (n = 225) in this figure.

type combinations were coherent in disease-state signatures, whereas only 25% of combinations were coherent in activity signatures (one-sided sign test, p = 0.022 and 0.99, respectively). Together, these analyses demonstrated better directional compatibility of the SLE risk allele's effect with disease-state signatures rather than activity signatures. Although this might be reasonable considering that most GWASs are based on case-control design, this finding implied the failure of current GWASs to capture the critical biology of SLE represented by disease-activity signatures.

While activity signature genes did not locate around the current SLE risk variants, we speculated that some of them might contribute to the disease risk by modulating the eQTL effects of risk alleles; we refer to the genes with modulating eQTL effects as proxy genes (prGenes; Figure 7C; STAR Methods; Zhernakova et al., 2017). Therefore, we finally sought to test whether disease-activity signature genes act as prGenes for risk alleles. In 115 SLE patients with genotype data available, we examined the influence of disease-activity signatures genes on the eQTL effects of risk alleles (we again utilized the above-mentioned colocalization results). Intriguingly, we detected two significant prGenes among activity signature genes (analysis of variance [ANOVA] test, FDR < 0.05; Table S6E), which included *MED24*, a transcriptional coactivator. *MED24* is an activity signature gene of plasmablasts (Figure 7F, left), and its expression was suppressed by MMF (Figure S7C). *MED24* significantly modulated the eQTL effect of an SLE risk variant (rs36059542) on *ARHGAP31*, encoding a GTPase-activating protein (ANOVA test, FDR = 0.035; Figure 7F, right). Of note, the eQTL effect of rs36059542 on *ARHGAP31* was observed only in plasmablasts (Ota et al., 2021). Therefore, in addition to disease-state genes, disease-activity genes may also contribute to genetic risk by modulating the eQTL effects of risk alleles. Furthermore, MMF might indirectly suppress SLE genetic risk by controlling the *MED24* expression.

DISCUSSION

In this study, we extensively examined dysregulated gene expression patterns of SLE by profiling 27 immune cell types from 136 SLE and 89 HC donors. The strategy we took in this bulk transcriptome study with a high cellular resolution aimed to achieve the better quality to capture the precise biology linked to a wide variety of patient conditions in this disease. In this approach, we successfully profiled two distinct categories of lupus-relevant signatures, that is, disease-state and disease-activity genes, and revealed the underappreciated cell-type-specific mechanisms driving these signatures, most of which were not identified in previous scRNA-seq studies (Arazi et al., 2019; Der et al., 2019; Nehar-Belaid et al., 2020). Moreover, we demonstrated the value of these signatures in multiple applications: transferability to independent transcriptome datasets, shared components with drug responses, and consistency with the genetic signals.

SLE is a chronic disease characterized by relapsing and remitting disease course. A critical but unsolved question is how the multicellular pathophysiology is different (or not different) between disease development and exacerbation stages. Most of the previous SLE transcriptome studies have failed to provide clear answers since they separately conducted case-control and/or intra-case analyses and have not directly compared both signatures in the same study with a high cellular resolution. Our comprehensive transcriptome data successfully clarified the multiple cell-type-specific immune-mediated molecular mechanism characteristic of disease-state and activity, which highlighted the distinct biology behind the establishment and exacerbation phases of this disease.

The current realistic goal of SLE management is to achieve remission or low disease activity (LDA), not a cure (Fanouriakis et al., 2019; Franklyn et al., 2016). From this perspective, the disease-activity signature, not the disease-state signature, has implications for the development of biomarkers of treatment response or therapeutic targets. Indeed, we observed shared components between the disease-activity signatures and transcriptome changes in BLM good responders and in those taking MMF. The cell-type-specific activity signatures identified in this study, for example, myeloid-lineage cells characteristic of renal disease, might be informative for designing novel treatment strategies for SLE.

We have provided in-depth integrative analyses of the risk variants and transcriptome signatures; these results have several important implications. First, since we recruited established SLE patients in the chronic phase, whether the disease-state signature genes indicate causality (the signatures induce disease state) or a reverse causality (the disease state induces the signatures) is a critical question. Since the risk variants reflect the causality, the fact that the risk variants are enriched around disease-state signature genes suggested the former scenario, further supported by the consistency in the dysregulation direction (i.e., coherent genes). Moreover, this finding also suggested that genetic risk-driven susceptibility signatures remained in clinically stable SLE patients, indicating that the causal mechanism is not completely controlled by the current treatments. Second, the current case-control GWAS signals failed to reflect the disease-activity signatures. This is a critical limitation of the current genetic studies considering the potential importance of activity signatures in drug target discovery. Although polygenic risk score based on such GWASs predicted poorer outcomes to some extent (Reid et al., 2020), a new framework of genetic study directly focusing on intra-case heterogeneity (e.g., disease severity) will be required.

It is now clear that disease-state and disease-activity signatures jointly maintain the complex pathophysiology of SLE. These signatures have the potential to be a pivotal foundation for future genomic, genetic, and drug discovery studies.

Limitations of the study

There are several limitations in our study. First, apart from the pre- and post-BLM subcohort, our study lacked longitudinal data. Second, all participants in this study were from the EAS population although we demonstrated the applicability of our signatures to transcriptome and GWAS data from multiple ancestries. Third, our cell-sorting strategy might have failed to characterize currently unidentified cell populations.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.cell.2022.07.021.

ACKNOWLEDGMENTS

The super-computing resource was provided by Human Genome Center, Institute of Medical Sciences, The University of Tokyo. This study was supported by Chugai Pharmaceutical Co., Ltd., Tokyo, Japan, the Ministry of Education, Culture, Sports, and the Japan Agency for Medical Research and Development (AMED) (JP21tm0424221 and JP21zf0127004). We thank all participants in this study. We thank all collaborators in the University of Tokyo, National Center for Global Health and Medicine, St. Luke's International Hospital, and Chugai Pharmaceutical for the contribution to sample collection and processing. M.N. thanks Michihiro Kono for helpful feedback. H.Suetsugu was supported by the Practical Research Project for Rare/Intractable Diseases from Japan AMED. X.Y. was supported by the Distinguished Young Scholar of Provincial Natural Science Foundation of Anhui (1808085J08). S.-C.B. was supported by the Basic Science Research Program through the National Research Foundation of Korea funded by the Ministry of Education (NRF-2021R1A6A1A03038899).

AUTHOR CONTRIBUTIONS

M.N. and K.I. designed the study. M.N. and K.I. wrote the manuscript with critical inputs from M.O. and K.F. M.N. conducted all analyses with the help of K.I., M.N., M.O., Y.T., Y.I., H.H., Y.N., T.I., J.M., R.Y., S.Y., A.N., HarukaTakahashi, HideyukiTakahashi, Y.A., and T.K.; and H.Shoda managed and contributed to sample collection of RNA-seq data. H.Suetsugu, L.L., K.K., X.Y., S.-Y.B., Y.C., H.-S.L., X.Z., S.-C.B., and C.T. managed EAS SLE-GWAS data. K.Y., T.O., and K.F. designed the project. All authors contributed to the final manuscript.

DECLARATION OF INTERESTS

M.O., Y.T., Y.N., and T.O. belong to the Social Cooperation Program, Department of Functional Genomics and Immunological Diseases, supported by Chugai Pharmaceutical. K.F. receives speaking fees, consulting honoraria, and research support from Chugai Pharmaceutical.

Received: January 26, 2022
Revised: May 19, 2022
Accepted: July 22, 2022
Published: August 22, 2022

SUPPORTING CITATIONS

The following references appear in the supplemented information: Ha et al., (2019), Kapellos et al., (2019), Maecker et al., (2012), Sanz et al., (2008).

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
See Table S1B for flow-cytometry antibodies	N/A	N/A
Critical commercial assays		
MACSxpress Neutrophil Isolation Kit, human	Miltenyi Biotec	Cat#130-104-434
RNeasy Mini Kits	QIAGEN	Cat#74104
MagMAX-96 Total RNA Isolation Kit	Thermo Fisher Scientific	Cat#AM1830
SMART-seq v4 Ultra Low Input RNA Kit for Sequencing	Takara Bio	Cat#634894
HiSeq SBS Kit v4	Illumina	Cat#FC-401-4003
NovaSeq 6000 S4 Reagent Kit v1.5	Illumina	Cat#20012866
QIAmp DNA Blood Midi Kit	QIAGEN	Cat#51185
TruSeq DNA PCR-Free Library prep kit	Illumina	Cat#20015963
Deposited data		
Gene expression data	This paper	National Bioscience Database Center (NBDC): JGAS000486
Human reference genome NCBI build 38, GRCh38	Genome ReferenceConsortium	https://www.ncbi.nlm.nih.gov/grc/human
Gene expression data for replication analysis	Panwar et al., 2021	GEO: GSE149050
Gene expression data for replication analysis	Takeshima et al., 2022	NBDC: hum0214.v3
Gene expression data for replication analysis	Andreoletti et al., 2021	GEO: GSE164457
Original codes	This paper	https://github.com/MasahiroNakano-hub/ImmuNexUT_SLE
MSigDB HALLMARK gene set collection	Liberzon et al., 2015	https://www.gsea-msigdb.org/gsea/msigdb/collections.jsp
MSigDB C3 all TF targets gene set collection	Kolmykov et al., 2021	https://www.gsea-msigdb.org/gsea/msigdb/collections.jsp
KEGG PATHWAY Database	Kanehisa et al., 2000	https://www.genome.jp/kegg/pathway.html
EUR SLE-GWAS summary statistics	Langefeld et al., 2017	https://www.ebi.ac.uk/gwas/studies/GCST007400
NHGRI-EBI GWAS Catalog	Buniello et al., 2019	https://www.ebi.ac.uk/gwas/
Open Targets Genetics	Ghoussaini et al., 2021	https://genetics.opentargets.org/
Software and algorithms		
STAR (v2.5.3)	Dobin et al., 2013	https://github.com/alexdobin/STAR
HTSeq (v0.11.2.)	Anders et al., 2015	https://github.com/htseq/htseq
bcl2fastq (v2.20.0)	Illumina	https://jp.support.illumina.com/sequencing/sequencing_software/bcl2fastq-conversion-software.html
R (v4.0.2)	N/A	https://www.r-project.org/
edgeR (v3.32.1)	Robinson et al., 2010	https://bioconductor.org/packages/release/bioc/html/edgeR.html
lme4 (v1.1-27.1)	Bates et al., 2015	https://cran.r-project.org/web/packages/lme4/index.html
uwot (v0.1.10)	McInnes et al., 2018	https://cran.r-project.org/web/packages/uwot/index.html
variancePartition (v1.20.0)	Hoffman and Schadt, 2016	https://bioconductor.org/packages/release/bioc/html/variancePartition.html
NicheNet (v1.1.0)	Browaeys et al., 2020	https://github.com/saeyslab/nichenetr
clusterProfiler (v3.18.1)	Yu et al., 2012	https://bioconductor.org/packages/release/bioc/html/clusterProfiler.html
LDSC (v1.0.1)	Finucane et al., 2018	https://github.com/bulik/ldsc

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Other		
Gene PC loading score list	This paper	Table S1E
Disease-state and activity signature genes	This paper	Tables S2A and S2B
Gene signatures related to therapeutic agents	This paper	Tables S5A and S5D
Biorender	N/A	https://biorender.com/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Keishi Fujio (fujiok-int@h.u-tokyo.ac.jp).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- RNA-seq data used in this study have been deposited at the National Bioscience Database Center (NBDC) Human Database (https://humandbs.biosciencedbc.jp/en/) (Dataset ID: JGAS000486) and are publicly available as of the date of publication. Accession numbers are listed in the key resources table.
- Original codes utilized in this study are available on GitHub (https://github.com/MasahiroNakano-hub/ImmuNexUT_SLE).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon reasonable request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

This study was approved by the Ethics Committees of the University of Tokyo. Written informed consent was obtained from all participants. We have complied with all of the relevant ethical regulations. Age and sex of all participants are deposited in NBDC with the same accession number as denoted above. Information related to the subjects are described in Method details.

METHOD DETAILS

Subjects

This study was approved by the Ethics Committees of the University of Tokyo (G10095). Written informed consent was obtained from all participants. Healthy volunteers were recruited at the Department of Allergy and Rheumatology at the University of Tokyo Hospital. SLE patients were recruited at the Department of Allergy and Rheumatology at the University of Tokyo Hospital, Division of Rheumatic Diseases at National Center for Global Health and Medicine, or Immuno-Rheumatology Center at St. Luke's International Hospital. All participants were from the EAS ancestry.

All SLE patients met the 1997 revised version of the American College of Rheumatology classification criteria (Hochberg, 1997). The exclusion criteria for the discovery dataset (including 136 SLE patients) were: i) active malignancies or infections, ii) use of more than 20mg prednisolone (PSL) daily or equivalent at enrollment, iii) receive of intravenous methylprednisolone pulse, cyclophosphamide, rituximab, or BLM within 12 months before enrollment.

In addition, we also collected 22 paired samples just before and six months after the additional therapy of BLM (i.e., pre- and post-BLM). Among the pre-BLM samples, 21 samples were included in the discovery dataset, and one was re-sampled due to the interval between initial enrollment and BLM induction. None of the post-BLM samples were included in the discovery dataset. Therefore, we recruited 248 donors in total: 136 unique SLE patients, one pre-BLM, 22 post-BLM patients, and 89 healthy volunteers.

Sample processing and sequencing

In this study, all samples were collected based on phase 2 protocol in the ImmuNexUT (Ota et al., 2021); 27 immune cell types were purified from peripheral blood of each donor (Table S1A) and then each sample was subject to RNA-seq. We first isolated PBMCs by density gradient separation with Ficoll-Paque (GE healthcare) immediately after the blood draw. Erythrocytes were lysed with Ammonium-Chloride-Potassium lysing buffer (Gibco), and non-specific binding was blocked with anti-human Fc-gamma receptor antibodies (Thermo Fisher Scientific). We next sorted PBMCs into 26 immune cell types with purity > 99% using a 14-color cell sorter BD FACSAria Fusion (BD Biosciences) with the aim of 5,000 cells per cell type per individual (Table S1A). All flow cytometry reagents

were listed in Table S1B. Our immune cell gating strategy for flow cytometry is based on the latest knowledge of the established cell subsets in each cell lineage (Table S1A). Sorted cells were lysed and stored at -80°C. Then, RNA for each sorted sample was extracted using MagMAX-96 Total RNA Isolation Kits (Thermo Fisher Scientific). Libraries for RNA-seq were prepared using SMART-seq v4 Ultra Low Input RNA Kit (Takara Bio). Neutrophils were purified using MACSxpress Neutrophil Isolation Kits human (Miltenyi Biotec) with the aim of 1×10^6 cells immediately after the blood draw, lysed and stored at -80°C, followed by RNA isolation with an RNeasy Mini Kits (QIAGEN) and library preparation with SMART-seq v4 Ultra Low Input RNA Kits (Takara Bio). All prepared libraries were sequenced on HiSeq2500 (6169 samples) or NovaSeq6000 (217 samples) (Illumina) to generate 100 or 150 base paired-end reads, respectively. Genomic DNA was isolated from peripheral blood using QIAmp DNA Blood Midi kit (QIAGEN). Libraries were prepared using TruSeq DNA PCR-Free Library prep kit (Illumina), followed by whole-genome sequencing (WGS). WGS was performed only for the samples from Japanese individuals. The details of WGS data processing were reported in our previous study (Ota et al., 2021).

QC of the expression data

Adaptor sequences were trimmed using Cutadapt (v1.16) with default parameters and reads containing low-quality bases (Phred quality score < 20 in > 20% of the bases) were removed. Reads were aligned against the GRCh38 reference sequence using STAR (v2.5.3) (Dobin et al., 2013) with the UCSC (provided in the iGenomes UCSC hg38 bundle; https://support.illumina.com/sequencing/sequencing_software/igenome.html). In STAR, we set the parameters as follows to improve the accuracy to identify splicing junctions: ''–alignSJDBoverhangMin 1 –sjdbScore 3 –scoreGap -2''. Expression was counted with HTSeq (v0.11.2) (Anders et al., 2015), adjusting the parameters to unstranded libraries as follows: ''–stranded=no''.

We applied multiple sample quality control (QC) steps to ensure high quality data. For QC and quantification, we used RNA-SeQC (v2.4.2) with default parameters (Graubert et al., 2021). From 6,413 sorted samples, one sample with uniquely mapped read rates < 80% or unique read counts < 5×10^6 were excluded as a low-quality sample (Table S1D). To identify outlier samples, we calculated Spearman's correlations of the expressions between two samples from the same cell type and then removed 21 samples with mean correlation coefficient < 0.9. In addition, to identify potentially swapped samples, we calculated the concordance rates between RNA-seq-based genotype and WGS-based genotype at the heterozygous loci and excluded five samples with concordance rate < 0.9.

We then filtered out low expression genes (<10 counts or <1 count per million [CPM] in > 85% of samples), followed by a trimmed mean of M values (TMM) normalization with R (v4.0.2) package edgeR (v3.32.1) (Robinson et al., 2010) in each cell type. Normalized expression data were converted to log-transformed count per million (i.e., log[CPM+1]). The batch effects (i.e., product lots in SMART-seq v4 and sequencer; Figure S1G) were removed using Combat software (Johnson et al., 2007). To verify the successful work of the batch correction procedure, we used principal variance component analysis; we first calculated the explained variance of each clinical parameter for each PC1-7 score with the linear mixed models in R package lme4 (v1.1-27.1) (Bates et al., 2015) and then inferred the average value of the explained variance weighted by each PC's eigenvalue (Figure S1H).

PCA and UMAP of all samples

For PCA and UMAP using all ImmuNexUT samples (Figures S1C and S1D), we combined the expression data after batch correction from each cell type and used the intersection of the genes (n=8,397) that passed the filtering of low expression in each cell type. We performed PCA using prcomp function in R (v4.0.2) after we centered and scaled the normalized expressions (i.e., log[CPM+1]) of these 8,397 genes. For UMAP, we used R package uwot (v 0.1.10) with default parameters (McInnes et al., 2018).

PCA in each cell type

For PCA in each cell type of the discovery cohort, we used the top 10,000 variable genes from the expression data after batch correction in each cell type (Figures 2B and S2A). For each cell type, we performed PCA using prcomp function in R after we centered and scaled the normalized expressions (i.e., log[CPM+1]) of the 10,000 genes.

IRG expressions in the ImmuNexUT

We evaluated the expression of 100 IRGs reported in a recent PBMC scRNA-seq study (Nehar-Belaid et al., 2020) across all 27 cell types and case-control. To recapitulate cell-type- and disease-specific IRG expression patterns described in the original publication, we first showed the expression levels of IRGs with the identical gene orders with the original publication (Figure S1I). Expression levels are scaled for each gene across all cell types. Cell types are arranged consistently with the original publication where applicable. Next, to discover novel cell-type-specific IRG expression patterns specific to the ImmuNexUT including neutrophil-lineage cells, we performed hierarchical clustering of IRGs and cell types based on the IRG expressions (Figure 2A). For the hierarchical clustering, Euclidean distances of the scaled expression levels were used with Ward's method and the cutoff for cluster tree of IRGs was set as Euclidean distances = 12.

The association of PCs with disease status

To evaluate how the transcriptome variation reflected the heterogeneous clinical statuses in the discovery dataset, we tested their associations with the PC scores for each cell type. This study focused on 14 clinical parameters related to disease-state, overall

disease-activity, organ/domain activity, and treatment statuses (Figure S2E). Disease-state was defined as the contrast between inactive SLE (i.e., not all SLE) and HC in the discovery dataset to exclude the elements of disease-activity signatures from the case-control contrast (Figure 1, middle). For overall disease activity, we defined four categories: i) inactive as SLEDAI-2K (Gladman et al., 2002) = 0, ii) low disease activity (LDA) as $1 \leq$ SLEDAI-2K ≤ 4, iii) moderate disease activity (MDA) as $5 \leq$ SLEDAI-2K ≤ 8, and iv) high disease activity (HDA) as SLEDAI-2K ≥ 9. For organ activity, we categorized the patients into seven groups based on their actively involved organ/domains of the British Isles Lupus Assessment Group (BILAG) 2004 (Isenberg et al., 2005) and SELDAI-2K: a) constitutional, b) mucocutaneous, c) musculoskeletal, d) renal, e) extrarenal severe (neuropsychiatric/eye, cardiorespiratory and/or gastrointestinal), f) hematological, and g) serological activities. We also evaluated the effect of therapeutic agents such as MMF, hydroxychloroquine (HCQ), and tacrolimus (TAC) (Durcan et al., 2019; Fanouriakis et al., 2019; Liu et al., 2015).

Since cumulative explained variance reached 50-60% until PC30 in all cell types, we tested the associations of the 14 clinical parameters and PC1-30 scores from each cell type using a linear regression model, adjusted for potential confounding factors (e.g., age, sex and treatments). We fitted the PC scores to the following linear regression models:

For disease-state (x: inactive SLE vs. HC),

$$y = \beta \cdot x[0,1] + \varepsilon \cdot Age[yrs] + \epsilon \cdot Sex[0,1] + \theta \quad \text{(Equation 1)}$$

Here, y represents the scaled PC score for each cell type (PC1-30 × 27 cell types). All PC scores were scaled across samples to enable the direct comparison of the effect sizes in the associations with clinical parameters. In this comparison, age and sex were included as covariates.

For disease-activity (x: HDA vs. inactive SLE; we also examined LDA vs. inactive and MDA vs. inactive SLE),

$$y = \beta \cdot x[0,1] + \sum_{k=1,2,3} \gamma \cdot I_k[0,1] + \delta \cdot PSL[mg] + \varepsilon \cdot Age[yrs] + \epsilon \cdot Sex[0,1] + \theta \quad \text{(Equation 2)}$$

Here, $I_k(k = 1,2,3)$ represents each immunosuppressant (MMF, HCQ, TAC) as covariates.

For organ/domain activity ($x_j[j = 1 \ldots 7]$: the abovementioned seven categories),

$$y = \sum_{j=1\ldots7} \beta \cdot x_j[0,1] + \sum_{k=1,2,3} \gamma \cdot I_k[0,1] + \delta \cdot PSL[mg] + \varepsilon \cdot Age[yrs] + \epsilon \cdot Sex[0,1] + \theta \quad \text{(Equation 3)}$$

Here, we constructed multiple linear regression models including all seven categories, which enabled us to infer the association of each organ activity with PC scores, controlling the other organs' effects (Figures 5C and 5D).

For therapeutic agents ($I_k[k = 1,2,3]$: MMF, HCQ, TAC),

$$y = \sum_{k=1,2,3} \gamma \cdot I_k[0,1] + \beta \cdot x[0,1,2,3] + \delta \cdot PSL[mg] + \varepsilon \cdot Age[yrs] + \epsilon \cdot Sex[0,1] + \theta \quad \text{(Equation 4)}$$

Here, x represents disease activity (inactive, LDA, MDA, and HDA) as covariates.

These equations enabled us to derive the associations of disease-state, activity, organ involvements or treatment statuses with PC scores, adjusted for other confounding factors. Statistical significance was set at FDR < 0.05 with the Benjamini-Hochberg method. All PC scores were signed so that the effect sizes of disease-state and disease-activity were positive (Figures 5A, 5B, S5A, and S5B). In the case of some PCs that had opposite sign in the effect sizes of disease-state and activity, the association with lower P value was prioritized to have positive effect size.

In total, 105 significant associations were identified (Table S4A). Most of the significant associations were detected within PC1-7, with larger numbers than average per PCs (Figure S2F). As a side note, we also observed that PC1-7 from each cell type (single-cell-type-PC) could efficiently capture the independent association signals rather than the PC scores using all sample across different cell types (all-cell-type-PCA) (Figure S2K). Therefore, we confirmed that PC1-7 from each cell type is a minimum set to associate the transcriptome with the clinical parameters in the discovery dataset (Figures S2G and S5A) and utilized PC1-7 scores for the subsequent analyses.

To calculate the proportion of sum squared deviations within HC, SLE, and between HC and SLE data in each cell type (Figure 2C), we used the PC1-7 data of the discovery dataset. We first calculated the proportion of sum squared deviations for each PC score and then inferred the average value of the proportion weighted by each PC's eigenvalue.

In the hierarchical clustering of 225 unique individuals using 189 PCs (= 7 PCs × 27 cell types) in the discovery dataset, the Euclidean distances of the PC scores were used with Ward's method (Figures 5A and S5A).

Weighted variance partitioning analysis

To calculate the explained variance of each factor for whole transcriptome variation, we fitted the each PC score to the linear mixed models in R package variancePartition (v1.20.0) (Hoffman and Schadt, 2016) and then inferred the average value of the explained variance weighted by each PC's eigenvalue for each cell type.

To calculate the explained variance of cell types and case-control status in all RNA-seq samples across cell types, we used all-cell-type PC1-2 data (Figures S1B, S1C, S1E, and S1F). To calculate the explained variance of case-control status for each cell type, we used each-cell-type PC1-2 data (Figures S2A–S2D). To calculate the explained variance of each clinical parameter within SLE

transcriptome data for each cell type, we used each-cell-type PC1-7 data of the discovery and external datasets (STAR Methods; the association of PCs with disease status; Figures 2D, 5C, S2L, S5E, S5H, and S6F).

For intra-SLE analyses in the discovery dataset, to verify whether the inferred explained variance was not biased by outlier samples, we estimated standard errors (S.E.) of the explained variance by jackknife resampling method (Figures 5C, S5E, and S6F). When we had *n* samples for one cell type, we re-calculated the explained variance *n* times by excluding each one of the samples. We then evaluated the distribution of *n* explained variance and quantified its S.E. For each clinical parameter, we compared the explained variance in each cell type against the median explained variance across all 27 cell types. To assess the significance of the difference observed in this comparison for a cell type, we utilized *n* explained variance calculated in jackknife resampling for that cell type; among *n* values, we calculated the proportion of the values which was smaller than the median value, and we defined this proportion as jackknife resampling *P* (P_{jk}). For each clinical parameter, if one cell type passed the Bonferroni-corrected $P_{jk} < 0.05$, we concluded that the clinical parameter significantly contributed to the within-SLE transcriptome variation in that cell type.

Differential gene expression analysis

To detect DEGs in each cell type, we fitted the TMM-normalized counts in the discovery dataset to the generalized linear models (GLM) with negative binomial distribution using edgeR (v3.32.1) (Robinson et al., 2010). The equations in these GLM models were consistent with those in the linear models as described in the STAR Methods; the association of PCs with disease status; we utilized the Equations 1, 2, 3, and 4. Additionally, we also considered the batch effects as covariates in this analysis since TMM-normalized counts were not corrected for batch effects (STAR Methods; QC of the expression data). These equations enabled us to derive DEGs related to our focused disease traits or treatment statuses, adjusted for other confounding factors (Figures 3A, 3B, 6E, S5F, and S6E). Statistical significance was set at FDR < 0.05 with the Benjamini-Hochberg method. We defined (1) "disease-state signature genes" as significant DEGs between inactive SLE and HC, and (2) "disease-activity signature genes" as significant DEGs between HDA and inactive SLE for each cell type (Figure 3A).

To evaluate the similarities and differences between disease-state and activity signature genes, we calculated Jaccard similarity indexes as the ratio of the shared genes with the concordant sign between the disease-state and activity signatures (Figure 3A; orange dots) over the union of these two signature genes (Figure 3A; orange + red + blue dots) for each cell type. Considering the biological significance, we did not regard the DEGs with the discordant sign between these two signatures as shared genes. Based on the proportion of DEGs and Jaccard index, we classified 27 cell types into three patterns (Figure 3B). We first defined the cell types with Jaccard index > 0.15 as shared pattern and then classified the other cell types into disease-state or disease-activity dominant patterns based on which signature genes were numerically predominant.

To estimate the false positive rates of edgeR for our disease-state and activity signature genes, we performed 1,000 permutation tests (Li et al., 2022). For each test, we randomly shuffled the sample label of our dataset and then performed differential gene expression analysis using edgeR (Figure S3E). We used the identical expression data as we used in the original analysis (we selected Naive CD4 samples in the discovery cohort as a representative data).

We also calculated the Jaccard similarity index and Spearman correlation across all pairs of different cell types for both signature genes. The Jaccard similarity distances (i.e., 1 - Jaccard similarity indexes) of each pair within disease-state signature genes were used for hierarchical clustering (Figure 3D). Similarly, Spearman's correlation distance of each pair within disease-activity signature genes were used for hierarchical clustering (Figure S3F).

To exclude the influence of one specific organ involvements on the results of activity DEG analysis, we performed the same activity DEG analyses excluding patients with one specific organ involvement (knock-out DEG analysis). Since there were only three HDA patients without serological activity, we did not apply this analysis to serological activity. Then, we compared logFC of each cell type from this knock-out DEG analysis with that from the original DEG analysis (27 cell types x 6 knock-out processes = 162 comparisons; Figure S5D).

To detect DEGs related to therapeutic agents adjusted for confounding factors (e.g., disease activity), we set the patients not taking the agent as the control, meaning the down-DEGs represented the genes that were downregulated by each agent. Therefore, we calculated Jaccard indexes as the ratio of the shared genes with the inverse sign between the disease-activity signatures and MMF-DEGs over the union of these two signature genes (Figure 6E). We calculated Jaccard indexes within the cell types in which more than 300 MMF-DEGs were observed.

Replication analysis

To assess the validity and generalizability of disease-state and activity signatures, we conducted replication analysis. We compared both signature genes with the immune cell RNA-seq data from the following three independent cohorts of SLE and/or HC from different ancestries. Importantly, the high cellular resolution is a big advantage of our discovery cohort, but at the same time it put hurdle on the replication analysis since many of the previous studies do not have the same-level cellular resolution as our study; hence this replication study is limited by imperfect cell-type matching (Table S2C).

i) Cohort 1 (Panwar et al., 2021).
   - Samples: 64 SLE and 24 HC, multi-ancestry cohort (Caucasian, Asian, Hispanic, and African).

- Cell subsets: six cell types (bulk T cells [T], bulk B cells [B], classical monocytes [CL Mono], myeloid DCs [mDC], plasmacytoid DCs [pDC] and Neu). Only CL Mono was collected from all donors, and the other five cell types were collected from around 20 SLE and 10 HC samples.
- Data usage strategy: since this cohort included both SLE and HC, we used this cohort for the replication analysis of disease-state and activity signatures (Figures S3G, S3H, and S5D). Naive CD4 and naive B cells (Naive B) data from the current study were compared with bulk T and bulk B data from cohort 1, respectively (Table S2C).

ii) Cohort 2 (Takeshima et al., 2022).
- Samples: 30 SLE and 37 HC, All East Asian (EAS).
- Cell subsets: 19 cell types (Naive CD4, memory CD4 [Mem CD4], Th1, T helper 2 cells [Th2], Th17, Tfh, fraction II effector regulatory T cells [Fr. II eTreg], Naive CD8, bulk memory CD8 [Mem CD8], NK, Naive B, USM B, SM B, DN B, plasmablasts, CL Mono, CD16 positive monocytes (CD16p Mono), mDC, and pDC).
- Data usage strategy: this is our previous cohort independent of the ImmuNexUT. In this study, we excluded the overlapped samples with the current study, leaving relatively stable 30 SLE patients for the analysis. Therefore, we used this cohort only for the replication of disease-state signatures (Figures S3G and S5C). Effector memory CD8 T cells (EM CD8) data from the current study was compared with bulk Mem CD8 data from cohort 2 (Table S2C).

iii) Cohort 3 (Andreoletti et al., 2021).
- Samples: 57 White and 63 Asian SLE patients.
- Cell subsets: four cell types (bulk CD4 T cells [CD4], NK cells, bulk B cells [B], and bulk monocytes [Mono]).
- Data usage strategy: since this cohort did not include HC samples, we used this cohort only for the replication of disease-activity signatures (Figures S3H and S5C). Naive CD4, Naive B, and CL Mono data from the current study were compared with bulk CD4, bulk B, and bulk monocyte data from cohort 3, respectively (Table S2C).

In all replications, we assessed the concordance of the directions of disease-state and activity signature genes in the discovery cohort with the corresponding genes in external cohorts using one-sided binomial sign tests. The definitions of clinical status (e.g., disease-state and activity) were consistent with our discovery cohort, with the exception that inactive SLE in cohort 2 was defined as $0 \leq$ SLEDAI-2K ≤ 2 (10 patients) since there were no patients with SLEDAI-2K = 0. In all replication analyses, we adjusted for the covariates (e.g., age and sex) in line with the analyses of the discovery dataset where applicable.

To assess the validity of disease-state and activity PCs in the discovery dataset, we projected the data from the three external datasets onto our PCA space in the corresponding cell types (Table S2C). In this PC projection method, we first computed the Z score matrix of gene expressions using the mean and standard deviation (SD) of the discovery dataset and then inferred the PC scores of each sample from the external datasets as the inner products of each PC loading (Table S1E) and the Z score matrix. We tested the association of these PC scores with disease-state (i.e., inactive SLE vs. HC) and disease-activity (i.e., HDA vs. inactive SLE) in the external datasets using the linear regression model as with the discovery dataset (Table S4B). We then assessed the concordance of the directions of the effect sizes for disease-state and disease-activity PCs, respectively, using one-sided binomial sign tests (Figure S5C).

To assess the contribution of clinical parameters to the within-SLE transcriptome variation in the replication cohorts, we again performed weighted variance partitioning analysis using replication cohort 1 and 3 (Figures S2L and S5H) as with the discovery cohort. We used PC1-7 from each cell type in line with the analysis in the discovery cohort. In Figure S2L, since cohort 3 did not provide the detailed clinical information other than disease activity, we could not calculate the variance explained by treatment, age and sex. Similarly, since cohort 1 did not provide the dose of PSL in most cases, we could not calculate the variance explained by PSL. In Figure S5H, we could not utilize cohort 3 since detailed information of organ activity was not available.

NicheNet analysis

To predict which cytokine ligand-receptor interactions could potentially induce the disease-state and activity signatures in each cell type, we performed NicheNet analysis (Browaeys et al., 2020) (Figures S4A and S4B). NicheNet predicts which ligands produced by a sender cell are the most active to induce gene expression in the receiver cells (upstream ligand activity analysis; Figures S4A and S4B, (1)). In this analysis, we considered the differentially expressed cytokine-encoding genes (i.e., disease-state and activity cytokines; Figure 4A) for which their corresponding receptors were expressed in at least one cell type (Figures S4A and S4B, (2)) as potential ligands. Ligand activity scores were inferred for the disease-state and activity signature genes of each cell type. In the ligand activity heatmap, we showed the union of top 3 prioritized cytokines for each signature. In the ligand-target genes heatmap (Figures S4A and S4B, (3)), we showed the union of the top 10 most strongly predicted targets of the prioritized cytokines. Among their regulatory potential scores, we omitted the lowest 20% scores for visualization of the heatmap, according to the developer's approach.

TF and pathway enrichment analysis

To test pathway and transcription factor (TF) enrichment in each disease-state and activity signature genes, we performed over-representation analyses with one-sided Fisher's exact test in R package clusterProfiler (v.3.18.1) (Yu et al., 2012). Statistical significance was set at FDR < 0.05 with the Benjamini-Hochberg method. For TF datasets, we used the Molecular Signatures Database (MsigDB)

C3 all TF targets annotation (1133 annotations) (Kolmykov et al., 2021). For pathway datasets, we used the MsigDB hallmark gene set collection (50 annotations) (Liberzon et al., 2015) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway (548 annotations) (Kanehisa and Goto, 2000). To capture the cell-type-specific biology linked to disease-state and activity signatures genes, we set the union of both signature genes in all cell types as the background gene sets. To overview the biology unique to each signature, we first classified all significant KEGG pathways to 33 subcategories according to the definition of the developers' website (https://www.genome.jp/kegg/pathway.html). We further classified these 33 subcategories to four main categories based on the biological relevance and summarized the number of enrichments for each category and subcategory, respectively (Figures 4C and S4E).

For organ activity-related and treatment-related DEGs (e.g., renal-DEGs, BLM-DEGs and MMF-DEGs), we restricted the pathway enrichment analysis to the cell types with the number of DEGs > 300.

Analysis of pre- and post-BLM dataset

All 22 individuals received BLM treatment according to the standard protocols (Navarra et al., 2011; Stohl et al., 2017). In this section, we defined those whose original disease-activity categories were moved into one or more lower categories (e.g., MDA to LDA or LDA to inactive) between pre- and post-BLM treatment, as good responders, and the others as poor responders.

Because edgeR did not implement generalized linear mixed models (GLMM), we detected DEGs between pre- and post-BLM using the following GLMM with negative binomial distribution in lme4 (v1.1-27.1) (Bates et al., 2015), setting the statistical significance at FDR < 0.05 with the Benjamini-Hochberg method (Figures 6A and 6B). Of note, we need to consider the batch effects as covariates in the following equation, since TMM-normalized counts were not corrected for batch effects (STAR Methods; QC of the expression data).

$$y = \beta \cdot BLM[0,1] + (1|individual) + \pi \cdot Batch + \theta$$

Here, y and BLM represents the TMM-normalized count for each gene in each cell type and the treatment status of BLM (i.e., pre- and post-BLM). We included a term for random intercept of individuals, and hence excluded individual-specific covariates (e.g., age and sex). In this equation, we set pre-BLM as the control, meaning the down-DEGs represented the genes that were downregulated by BLM treatment. Therefore, we calculated Jaccard indexes as the ratio of the shared genes with the inverse sign between the disease-activity signatures and BLM-related DEGs over the union of these two signature genes (Figure 6B). We calculated Jaccard indexes only in the B-lineage cells in which more than 300 BLM-DEGs were observed. To evaluate BLM effects on all cell types, we next extended this analysis to all activity signature genes, not restricting it to BLM-DEGs. We tested the associations between logFCs in disease-activity effects and those in BLM effects using linear regression model (Figures S6A and S6B).

To explain the difference between good-responder BLM-DEGs and poor-responder BLM-DEGs of B-lineage cells in Figure 6B, we additionally constructed the following GLMM, using all BLM-related samples (joint model; n=22x2):

$$Full: Count = \beta \cdot BLM[0,1] + \gamma \cdot response[0,1] + \delta \cdot response:BLM + (1|individual) + \pi \cdot Batch + \theta$$

$$Null: Count = \beta \cdot BLM[0,1] + \gamma \cdot response[0,1] + (1|individual) + \pi \cdot Batch + \theta$$

This joint model enabled us to directly examine treatment response-specific BLM-DEGs (as indicated by the interaction term "response: BLM" in the model), adjusting for individual-level effect as a random effect (Table S5B). In this model, the expected behavior of the effect sizes for the treatment response-specific BLM-DEGs (δ in the above model) would be as follows:

i) up-regulated BLM-DEGs of good responder in our original manuscript: $\delta > 0$.
ii) down-regulated BLM-DEGs of good responder in our original manuscript: $\delta < 0$.
iii) up-regulated BLM-DEGs of poor responder in our original manuscript: $\delta < 0$.
iv) down-regulated BLM-DEGs of poor responder in our original manuscript: $\delta > 0$.

We calculated the concordance rates of good- or poor-responder BLM-DEGs (up or down) in Figure 6B that showed expected signs of interaction effect size in the joint model (Figure S6C).

In the PC projection method, we first computed the Z score matrix of gene expressions using the mean and SD of the discovery dataset and then inferred the PC scores of duplicated samples as the inner products of each PC loading (Table S1E) and the Z score matrix. To test the association between the change of PC scores and the treatment status, we used the following linear mixed model:

$$y = \beta \cdot BLM[0,1] + (1|individual) + \theta$$

Here, y represents the scaled PC score for each cell type. In this model, it was not necessary to consider the batch effects since the PCA was performed using the log(CPM+1) gene expression data that had been already corrected for batch effects (STAR Methods; QC of the expression data).

Stratified LD score regression analysis

To evaluate the enrichment of the genome-wide distribution of all SLE risk variants irrespective of their effect sizes (heritability) around HC-SEG, disease-state and activity signature genes, we performed stratified linkage disequilibrium score regression (S-LDSC)

(Finucane et al., 2015, 2018). We examined the enrichment of SLE heritability for common variants within 100-kb windows on either side of the transcription start site of the genes with the top 1,000 highest Z-scores in either signature genes for each cell type, adjusting for baseline model provided by the developers (Finucane et al., 2015; Figure 7A). For this analysis, we used two large-scale SLE GWAS summary statistics from EAS (Yin et al., 2021) and EUR ancestries (Langefeld et al., 2017; Figure S7A). Since the regression coefficients of S-LDSC are influenced by the GWAS heritability, we normalized coefficients by dividing them with mean per-SNP heritability as reported by a previous report (Finucane et al., 2018); we then reported normalized coefficients. In a fixed-effect meta-analysis of the two results, we used the inverse variance weighting method using normalized coefficients and their S.E. We reported *P* values to test whether the regression coefficient is significantly positive.

To call HC-SEG (specifically expressed genes in HC) for each cell type, we compared the expression data of one cell type with that of the remaining cell types that belong to other cell lineages using the GLM with negative binomial distribution in edgeR (v3.32.1) (Robinson et al., 2010). To be in line with previous studies of S-LDSC, only the samples from HC were used in this analysis.

GWAS prioritized gene enrichment analysis

The SLE-GWAS results were downloaded from the NHGRI-EBI GWAS Catalog (Buniello et al., 2019) on 16/08/2021. To assign the most likely causal genes to SLE-GWAS significant variants ($P < 5 \times 10^{-8}$), we used the L2G score from Open Targets Genetics (Ghoussaini et al., 2021). We defined the genes that showed L2G score > 0.5 (Mountjoy et al., 2021) for at least one SLE-GWAS variant as SLE-GWAS prioritized genes (131 genes in total; Table S6B). Gene symbols were based on UCSC definition. To test the enrichment of SLE-GWAS prioritized genes for disease-state and activity signature genes, we performed over-representation analyses with one-sided Fisher's exact test (Figure 7B). We set the union of the genes that passed the filtering of low expression in each cell type and used it as the background.

Integrative analysis with eQTL data

To compare the direction between the risk allele's expression quantitative trait loci (eQTL) effects and disease-state and activity signature genes, we utilized the results of the colocalization test between SLE-GWAS and ImmuNexUT eQTL data reported in Ota et al. (2021). For visualization, logFC sign information was adjusted so that the direction of the coherent genes, which showed the concordant direction between eQTL effects for risk alleles and differential expressions (Marigorta et al., 2017; Figure 7E), was positive (i.e., adjusted logFC).

If a gene modulates the magnitude of eQTL effect on another eGene depending on its expression level, we refer to that gene as proxy gene (prGene) (Zhernakova et al., 2017). To examine the interactive effects of the signature genes on the eQTL effects of SLE risk variants as prGenes, we fitted the eGene expressions to the following linear regression models for each cell type (Figure 7F):

$$Full:\ y = \rho \cdot G[0,1,2] + \mu \cdot x + \beta \cdot G:x + \sum_{k=1,2,3} \gamma \cdot I_k[0,1] + \delta \cdot PSL[mg] + \varepsilon \cdot Age[yrs] + \epsilon \cdot Sex[0,1] + \theta$$

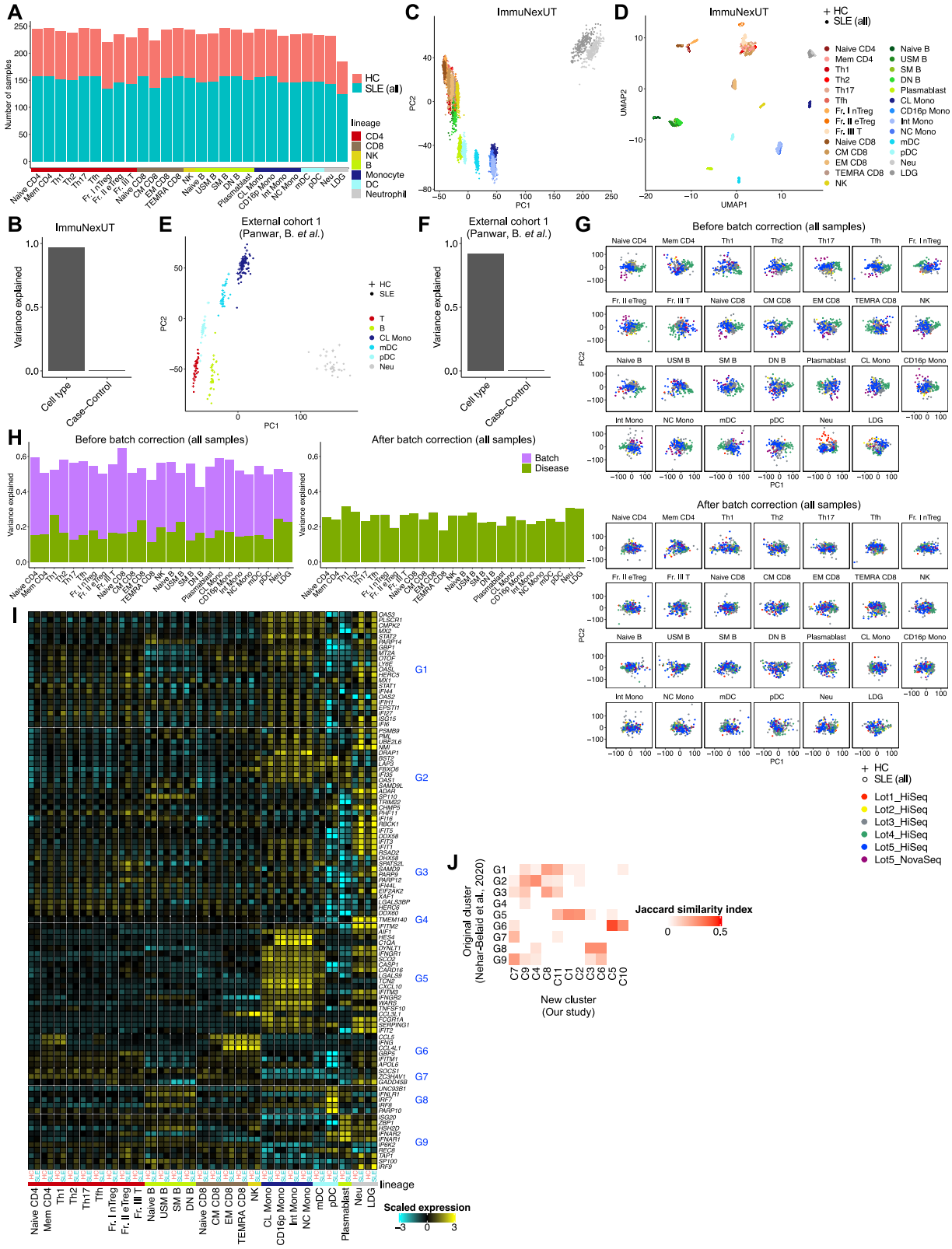
$$Null:\ y = \rho \cdot G[0,1,2] + \mu \cdot x + \sum_{k=1,2,3} \gamma \cdot I_k[0,1] + \delta \cdot PSL[mg] + \varepsilon \cdot Age[yrs] + \epsilon \cdot Sex[0,1] + \theta$$

Here, y and x represents the expression of eGene and prGene, respectively, and G represents the genotype of each individual. $I_k(k = 1,2,3)$ represents each immunosuppressant (MMF, HCQ, TAC) as covariates. We tested the significance of interaction terms (i.e., $G:x$) by comparing full and null models using analysis of variance (ANOVA). Statistical significance was set at FDR < 0.05 with the Benjamini-Hochberg method.

QUANTIFICATION AND STATISTICAL ANALYSIS

The statistical details and tests are indicated in the figure legends or STAR Methods. Throughout the analyses, multiple test correction was performed with Benjamini-Hochberg method unless otherwise indicated.

Supplemental figures



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Figure S1. Overview of gene expression patterns in the ImmuNexUT cohort, related to Figure 2

(A) A bar plot showing the number of samples that passed quality control in each cell type.

(B) A bar plot showing the proportion of variance explained by cell type and case-control in all ImmuNexUT data (6,386 samples from 248 donors).

(C and D) (C) A PCA and (D) a UMAP plot of all ImmuNexUT data across all cell types.

(E) A PCA plot of all data in the external cohort 1 as replication. In (C)–(E), colors and shapes indicate cell types and diseases, respectively.

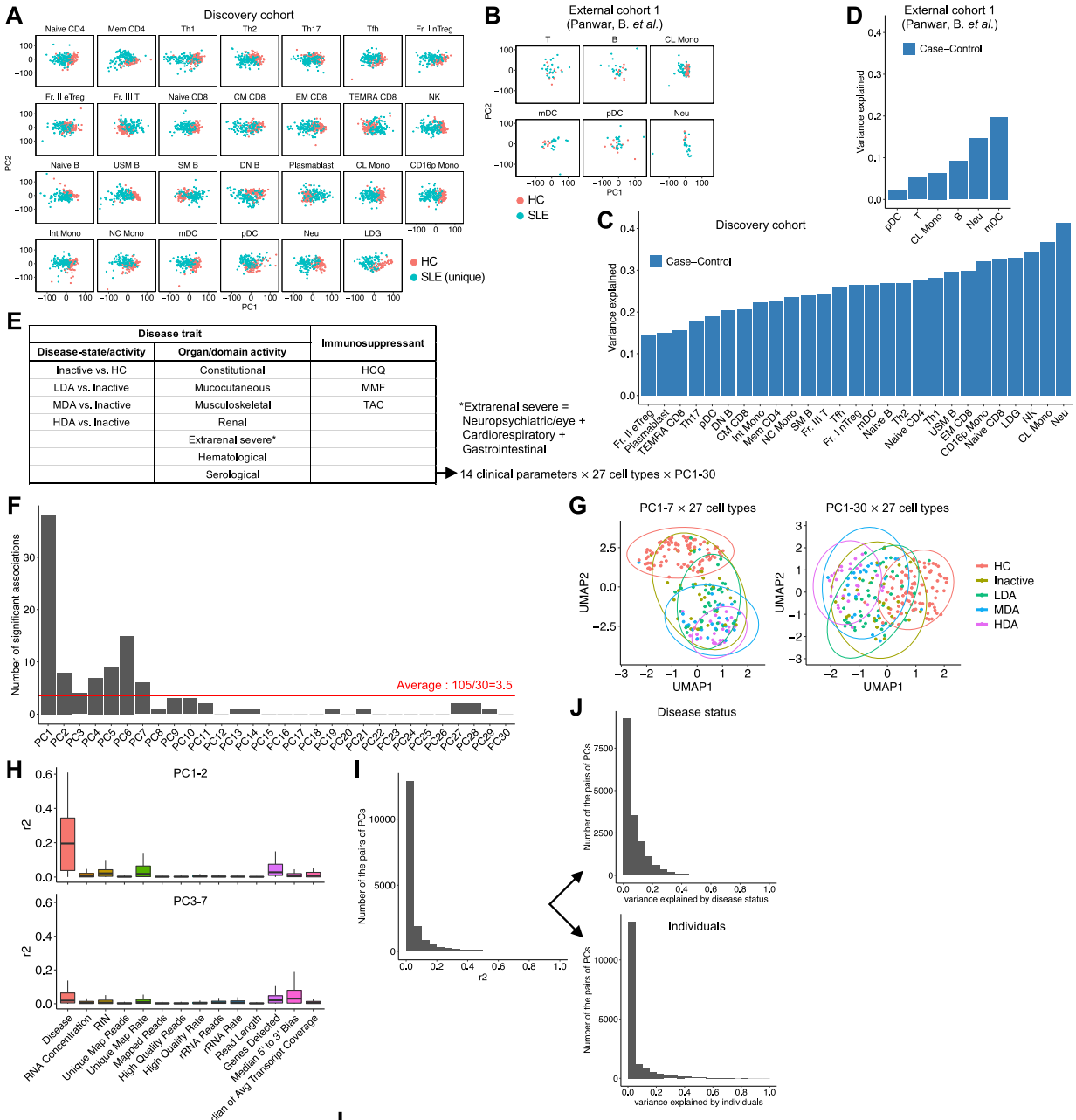
(F) A bar plot showing the proportion of variance explained by cell type and case-control in the external cohort 1.

(G) PCA plots of all ImmuNexUT data for each cell type (top) before and (bottom) after batch correction. Colors and shapes represent each batch and disease, respectively.

(H) Bar plots showing the proportion of variance explained by batch effect and disease in the expression data for each cell type (left) before and (right) after batch correction.

(I) A heatmap showing the mean expression of 100 IRGs across all cell types and diseases. Expression levels are scaled for each gene. Genes and clusters (G1–9) originate from a PBMC scRNA-seq study (Nehar-Belaid et al., 2020). Cell types are arranged consistently with the original publication where applicable. Several consistent cell-type- and disease-specific IRG expression patterns between our data and the original publication are shown: for example, upregulated *CXCL10* and *IFITM3* expression (G5) in lupus CD16p Mono and upregulated *IRF7* and *PARP10* expression (G8) in pDC.

(J) A heatmap showing the Jaccard similarity index of IRGs between the original clusters (G1–9) and the new clusters of the ImmuNexUT data (C1–11, Figure 2A). Column order is based on the hierarchical clustering using the Jaccard indexes. Some different IRG cluster patterns between our data and original publication are shown: for example, most of the original G5 genes (highest expression in DCs and monocytes in their dataset) are divided into C1 (highest in mDC, monocyte- and neutrophil-lineage cells), C2 (highest only in monocyte-lineage cells), and C11 (highest only in neutrophil-lineage cells).



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Figure S2. Rationale and strategies of PCA in this study, related to Figure 2

(A and B) PCA plots of HC and unique SLE data for each cell type in the (A) ImmuNexUT and (B) external cohort 1.

(C and D) Bar plots showing the proportion of variance explained by case-control in the (C) ImmuNexUT and (D) external cohort 1 data for each cell type. The larger variance explained in the ImmuNexUT (on average 25.8%) than in external cohort 1 (on average 9.5%) indicates the better case-control separation in the ImmuNexUT.

(E) A list of the 14 clinical parameters examined in this study. The association of these parameters and the PC scores in each cell type are tested in the linear regression model, adjusted for confounding factors.

(F) A histogram showing the number of significant associations between PC scores in each cell type and the clinical parameters (FDR < 0.05 in the linear regression test). The red horizontal line indicates the average number of significant associations per PC. Most of the significant associations (105 in total) are detected within PC1–7, with larger numbers than average per PCs.

(G) A UMAP plot of 225 individuals in the discovery dataset using (left) PC1–7 and (right) PC1–30 scores in all cell types. Colors represent the clinical status of individuals. UMAP using PC1–7 scores clearly distinguish clinically active SLE patients from HC.

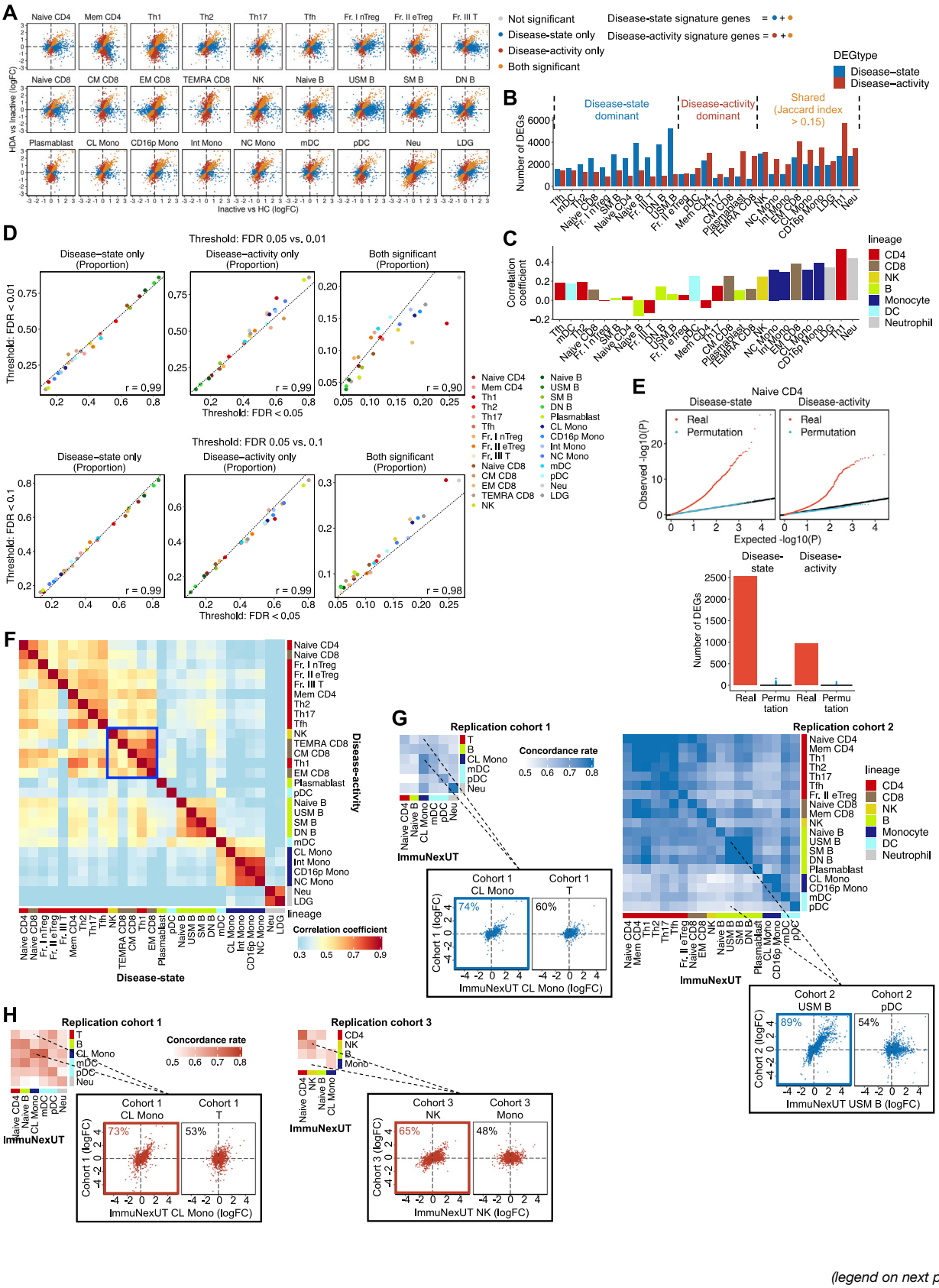
(H) Boxplots showing the correlations (r2) of PC1–2 (top) and PC3–7 (bottom) scores in all cell types with the technical covariates of RNA-seq as well as with case-control status ("disease"). PC1–7 scores in all cell types are not strongly correlated with the technical covariates.

(I) A histogram showing the correlations (r2) across all pairs of PC1–7 from all cell types. While there are on average low correlations (median r2 = 0.014), a few pairs of PCs show high correlations.

(J) A histogram showing the proportion of variance explained by disease status (top) and individuals term (bottom) for all pairs of PC1–7 from all cell types. Larger explained variance by disease status (median 0.043) than individuals (median 0) indicate that some high correlations in (I) reflect the disease status rather than the individual-specific effects (e.g., age, sex, and technical factors) unrelated to disease status.

(K) Bar plots showing the number of cell types that showed a significant association of single-cell-type and all-cell-type PC scores with the disease state (blue) and disease activity (red). Single-cell-type-PCA shows more association signals rather than all-cell-type-PCA. We analyzed the discovery dataset (n = 225) in (A)–(K).

(L) Bar plots showing the proportion of variance explained by the clinical parameters within-SLE data in each cell type of the external cohorts 1 (left) and 3 (right). On average, there are only 1.3% (cohort 1) and 1.2% (cohort 3) of the variance explained by disease activity, much smaller than those in our cohort (Figure 2D), probably due to the small numbers of clinically active patients in replication cohorts (six [9.4%] and four [3.3%] HDA patients in cohorts 1 and 3, respectively).



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Figure S3. SLE disease-state and disease-activity signatures, related to Figure 3

(A) Scatter plots comparing the logFC of disease-state and disease-activity signature genes in each cell type. Each dot represents one gene, colored based on the significance (FDR < 0.05 in differential expression test) of each comparison. Genes with logFC > 3 are plotted at the position of logFC = 3.

(B) A barplot showing the numbers of disease-state and activity signature genes in each cell type.

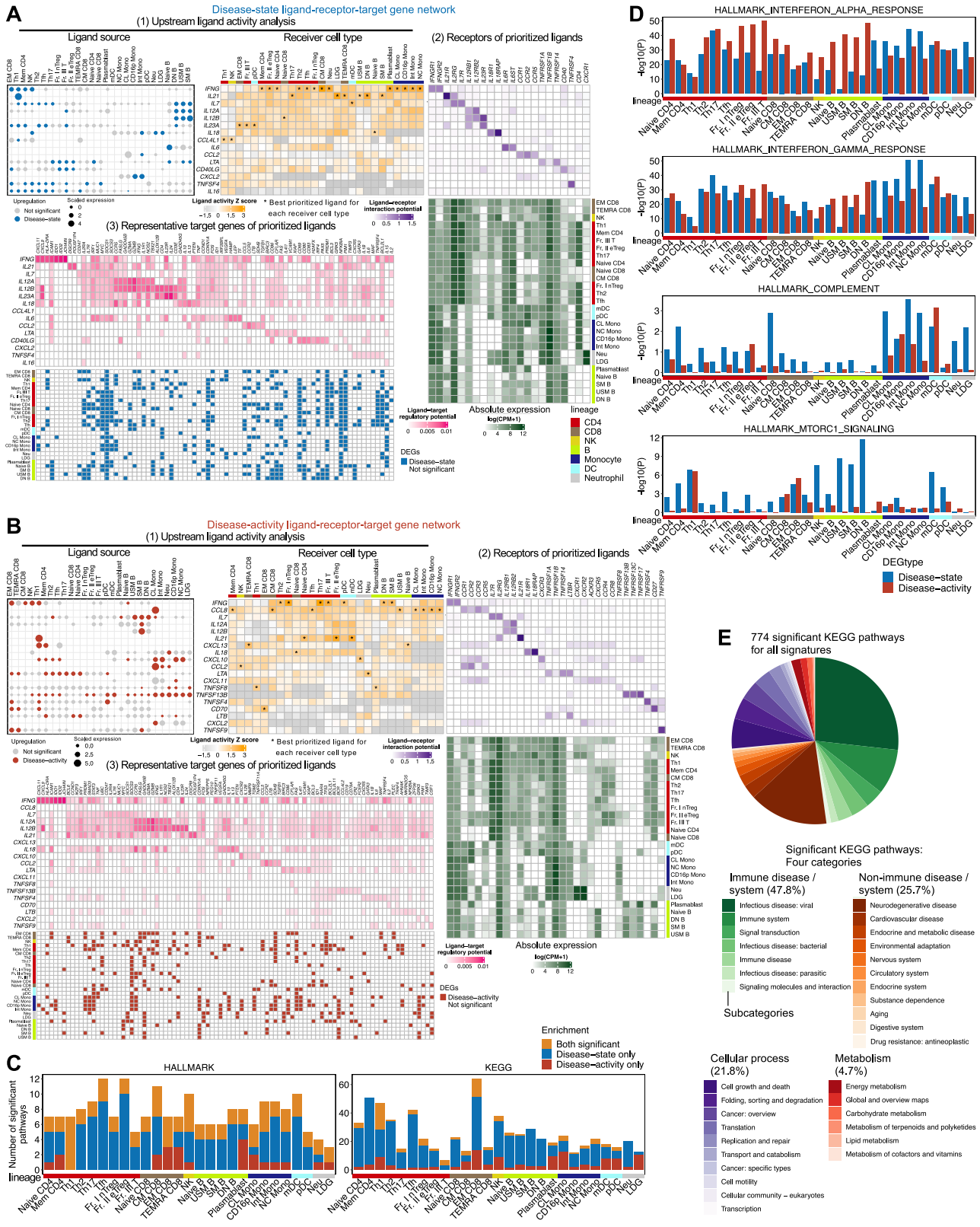
(C) A barplot showing the Spearman correlations between the logFC in disease-state and disease-activity signatures for each cell type. In (B) and (C), cell types are separated into three groups.

(D) Scatter plots comparing the proportion of three DEG types (disease-state only, disease-activity only, and both significant DEGs) in each cell type among different significance thresholds for DEGs. Pearson correlation coefficient (r) is shown at the bottom right in each comparison. The thresholds at FDR < 0.01 and FDR < 0.1 show consistent patterns with those at FDR < 0.05: the proportions of three DEG types show high correlations (0.90–0.99).

(E) (Top) Quantile-quantile plots comparing the observed p values of differential gene expression analysis (left; disease-state, right; disease-activity) between in the real data (red) and one representative permutation data (blue) of naive CD4 samples. (bottom) Bar plots and boxplots showing the number of DEGs (FDR < 0.05, left; disease-state, right; disease-activity) in the real data (red bar plots) and 1,000 times of permutation data (blue boxplots) of naive CD4 samples (see also STAR Methods). In the permutation data, p values are well controlled and false-positive DEGs are barely observed both for disease-state and disease-activity DEGs: median (IQR) number of DEGs in the 1,000 permutation tests is 0 (0-1) for disease-state and 0 (0-0) for disease-activity signatures.

(F) A heatmap showing the Spearman correlations across all cell types in disease-state and disease-activity signatures. The order of cell types in row and column is the same and based on the hierarchical clustering using the Spearman correlation coefficients of disease-activity signatures. Row and column annotation colors indicate cell lineages. We analyzed the discovery dataset (n = 225) in (A)–(F).

(G and H) (Top) Heatmaps showing the concordance rates (G) between the disease-state signatures in the current study and those in replication cohort 1 (left; HC, n = 24; inactive SLE, n = 16) and 2 (right; HC, n = 37; inactive SLE, n = 10) and (H) between the disease-activity signatures in the current study and those in replication cohort 1 (left; inactive, n = 16; HDA SLE, n = 6) and 3 (right; inactive, n = 41; HDA SLE, n = 4). Good concordance in disease-state signatures (one-sided sign test, $p < 10^{-34}$ and $p < 10^{-76}$ for each pair of corresponding cell types from cohorts 1 and 2, respectively) and in disease-activity signatures (one-sided sign test, $p < 10^{-8}$ and $p < 10^{-3}$ for each pair of corresponding cell types from cohorts 1 and 3, respectively) are observed. (Bottom) Scatter plots comparing (G) disease-state logFC of the current study and those of replication cohorts 1 and 3, and (H) disease-activity logFC of the current study and those of replication cohorts 1 and 2 in the representative corresponding (left plots in blue or red frame) and noncorresponding (right plots in black frame) cell-type combinations. Concordance rate is shown at the upper left for each comparison.



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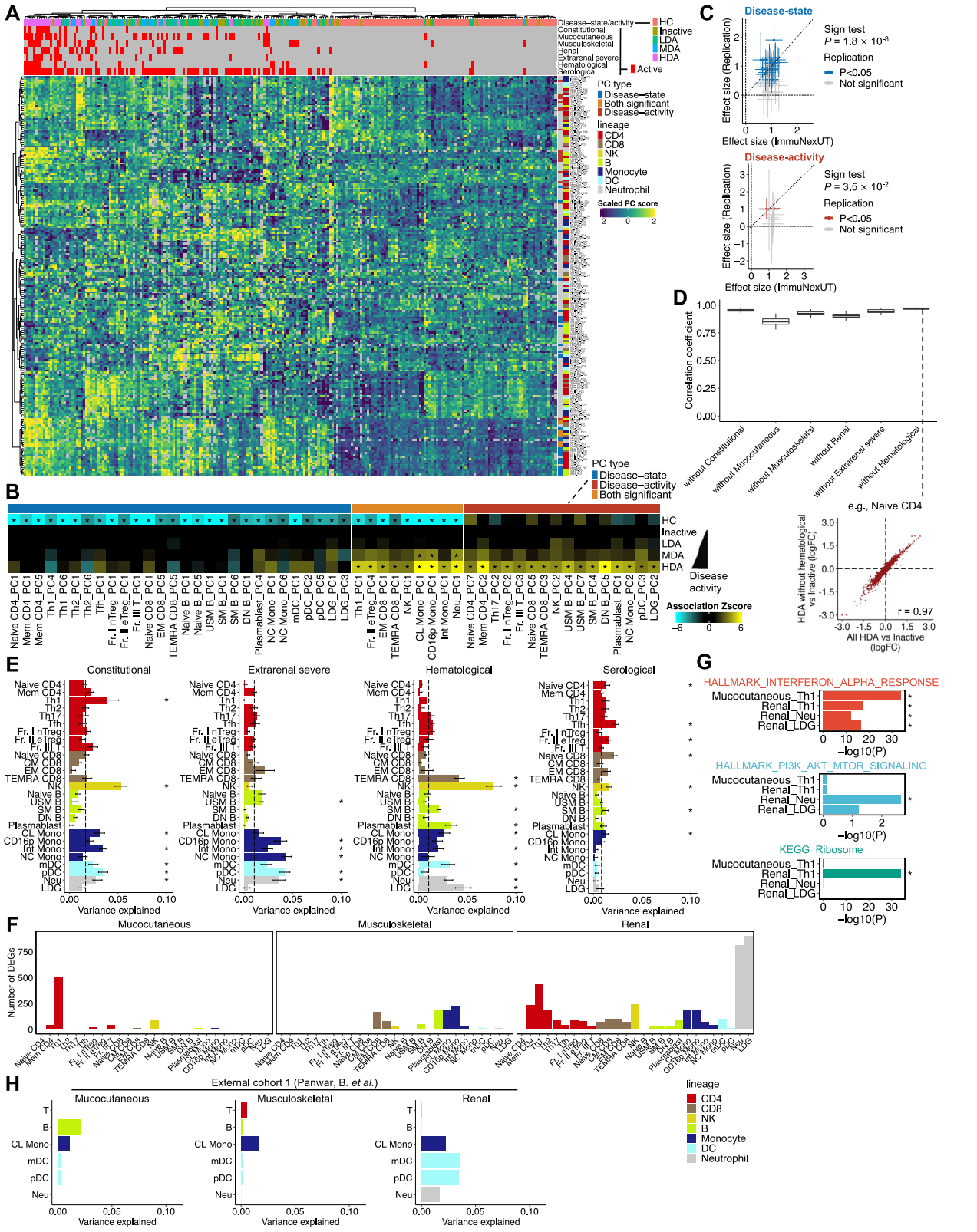
Figure S4. Cell-type-specific biology in disease establishment and exacerbation, related to Figure 4

(A and B) (A) Disease-state and (B) disease-activity ligand-receptor-target gene network analysis using NicheNet (see also STAR Methods). In panel (1), the differential expression (left) and ligand activity scores (right) of the union of the top 3 prioritized cytokines for each signature are shown. In panel (2), the absolute expression of the receptors that correspond to prioritized ligands (bottom) and their interaction potential scores (top) are shown. In panel (3), the regulatory potential scores between the prioritized cytokines and the union of the top 10 most strongly predicted targets (top) and the differential expression status of the target genes (bottom) are shown. For disease-activity genes, *IFNG* and *CCL8* are ranked as the top in seven and eight receiver cell types, respectively (B, 1). Only CL mono expresses *CCL8* as an activity signature gene (B, 1), and multiple cell types express its corresponding receptors (*CCR1*, 2, 3, and 5) (B, 2). These findings suggest that *CCL8* from CL mono and its receptors network could play an important role in SLE exacerbation.

(C) Bar plots showing the number of significant (left) MsigDB HALLMARK and (right) KEGG pathway enrichments for disease-state and disease-activity signature genes in each cell type.

(D) Bar plots showing the enrichments of representative MsigDB HALLMARK pathways for each signature. p, p values in one-sided Fisher's exact test. The strong enrichments of established pathways for SLE are shown: for example, IFN response in all cell types for both signatures (Crow, 2014), complement activation in monocyte-lineage cells and mDC for both signatures (Leffler et al., 2014), and mammalian target of rapamycin complex (mTORC) 1 signaling in lymphocytes and DC-lineage cells predominantly for disease-state signatures (Perl, 2016).

(E) Classification of 774 KEGG pathway annotations significantly enriched in any signatures (FDR < 0.05 in one-sided Fisher's exact test) into four main categories and 33 subcategories. We analyzed the discovery dataset (n = 225) in this figure.



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Figure S5. SLE disease-state and disease-activity PCs, related to Figure 5

(A) Hierarchical clustering of 225 unique individuals based on all PC1–7 scores of 27 cell types. Top annotations indicate the disease status and organ/domain activities in each individual. Right annotations indicate the type and cell lineage of each PC.

(B) A heatmap showing the association of PC scores with disease-state/-activity in linear regression test. All PCs with significant association with disease state and/or disease activity are shown (*FDR < 0.05).

(C) Scatter plots comparing the effect sizes of (top) disease-state PCs and (bottom) disease-activity PCs in the current study and those in replication cohorts (cohorts 1 and 2 for disease state and cohorts 1 and 3 for disease activity). The PCs with nominal $p < 0.05$ in linear regression test in replication cohorts are colored. Error bars indicate 95% confidence intervals. For disease-state PCs, 94% (33/35) of the PCs show concordant sign between the current study and replication cohorts (one-sided sign test, $p = 1.8 \times 10^{-8}$). For disease-activity PCs, 88% (7/8) of the PCs show concordant sign between the current study and replication cohorts (one-sided sign test, $p = 3.5 \times 10^{-2}$).

(D) (Top) A box plot showing the Pearson correlation coefficients between the logFC comparing all HDA versus inactive SLE and the logFC comparing HDA without one specific organ activity versus inactive SLE for each cell type (knockout DEG analysis; see also STAR Methods). Overall, there are high correlations of logFC among all comparisons (the medians of Pearson correlation coefficients [r] across 27 cell types in each organ knockout process = 0.85–0.97), indicating that overall disease activity in our dataset is unlikely to be biased by one specific organ activity. (Bottom) A scatter plot showing a representative example of comparison (without hematological activity in naive CD4).

(E) Bar plots showing the proportion of variance explained by organ activities within-SLE data in each cell type. The results for the four organ/domain activities other than those in Figure 5C are shown. Error bars and dashed vertical lines indicate 95% confidence intervals from jackknife resampling and the median values across 27 cell types, respectively. *Bonferroni-adjusted $p_{jk} < 0.05$.

(F) Bar plots showing the numbers of DEGs in each cell type between patients with and without (left) mucocutaneous, (middle) musculoskeletal, and (right) renal activity. Consistent with the variance partitioning analyses (Figure 5C), DEGs are observed especially in Th1 for mucocutaneous, monocyte-lineage cells for musculoskeletal, and neutrophil-lineage cells for renal activity, respectively.

(G) Bar plots showing the enrichment of representative pathways for the organ-DEGs in mucocutanepus-Th1, renal-Th1, renal-Neu, and renal-LDG. p, nominal p values; *FDR < 0.05 in one-sided Fisher's exact test. Interferon-related pathways are enriched in all DEG lists. For renal-neutrophil DEGs, PI3K/AKT/mTOR signaling is enriched. We analyzed the discovery dataset (n = 225) in (A)–(G).

(H) Bar plots showing the proportion of variance explained by (left) mucocutaneous, (middle) musculoskeletal, and (right) renal activity within-SLE data in each cell type of the external cohort 1. On average, the variance explained by each organ activity is smaller than those in our cohort (Figure 5C), probably due to the small numbers of clinically active (six [9.4%] HDA) patients in replication cohort 1.

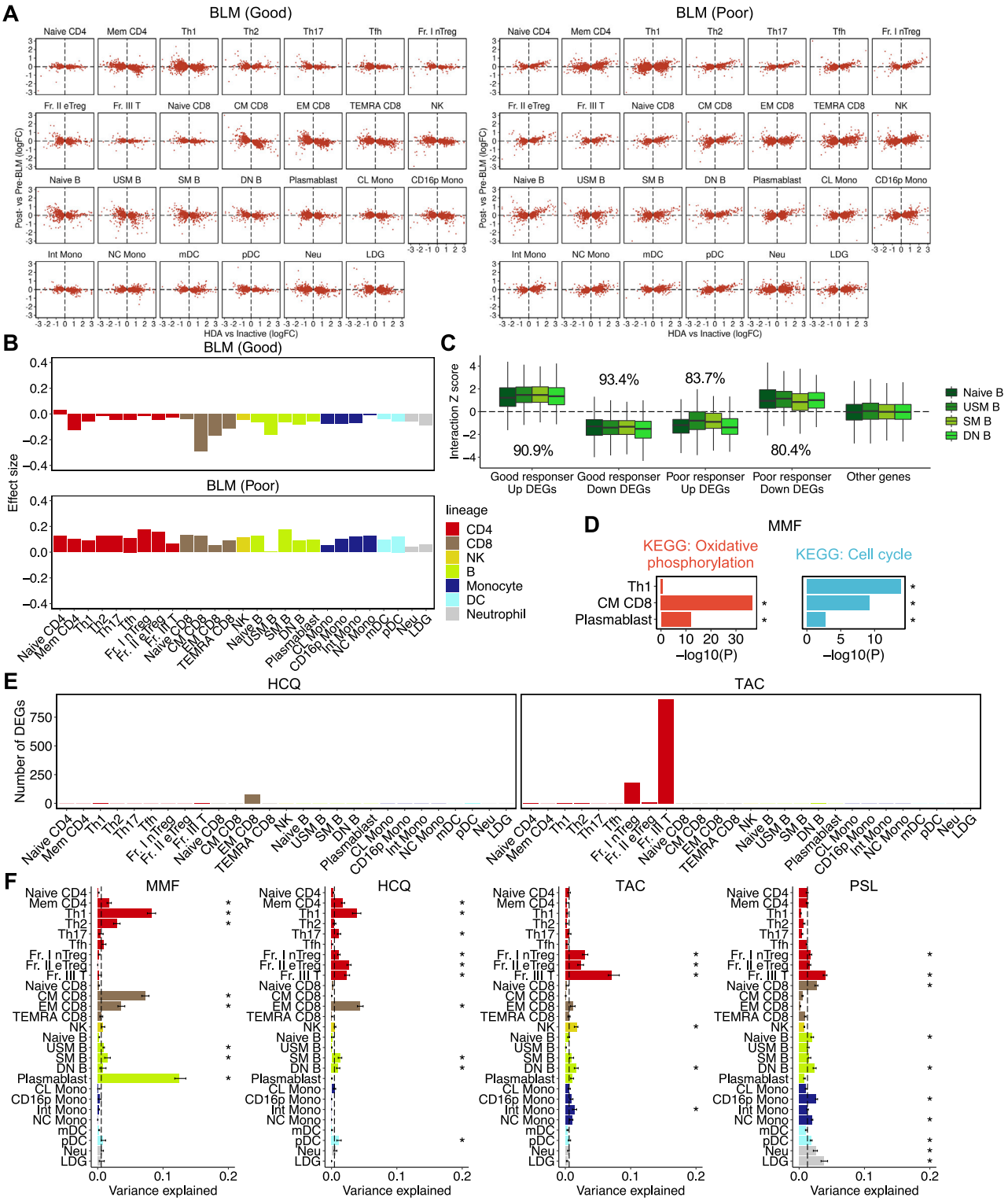


Figure S6. Cell-type-specific activity signatures linked to treatment responses, related to Figure 6

(A) Scatter plots comparing the disease-activity effects (logFC) and BLM effects (logFC) on the activity signature genes in each cell type, separated into good (n = 9) and poor responders (n = 13).

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(B) Bar plots showing the effect sizes in the linear regression tests for the association between disease-activity effects and BLM effects, separated into good and poor responders. Disease-activity signature effects, especially in B- and memory CD8-lineage cells, are negatively correlated with the BLM effects only in good responders (linear regression test; FDR < 0.05), suggesting BLM also modestly affects non-B-cell transcriptome.
(C) Box plots showing the *Z* scores of the interaction term (response: BLM) in the joint generalized linear mixed model using all BLM-related samples for each cell type (see also STAR Methods and Table S5). High concordance rates of good- or poor-responder BLM-DEGs (up or down) in Figure 6B that showed expected signs of interaction *Z* score are shown (80.4%–93.4% for each cell type, one-sided sign test $p < 1.3 \times 10^{-44}$).
(D) Bar plots showing the enrichment of representative pathways for the MMF-DEGs in Th1, CM CD8, and plasmablast. p, nominal p values; *FDR < 0.05 in one-sided Fisher's exact test.
(E) Bar plots showing the numbers of DEGs between patients with versus without (left) hydroxychloroquine (HCQ) and (right) tacrolimus (TAC) in each cell type.
(F) Bar plots showing the proportion of variance explained by medication status of MMF, HCQ, TAC, and PSL within-SLE data in each cell type. Error bars and dashed vertical lines indicate 95% confidence intervals from jackknife resampling and the median values across 27 cell types, respectively. *Bonferroni-adjusted $p_{jk} < 0.05$. For the analyses other than BLM-related analyses, we used SLE data in the discovery dataset (n = 136).

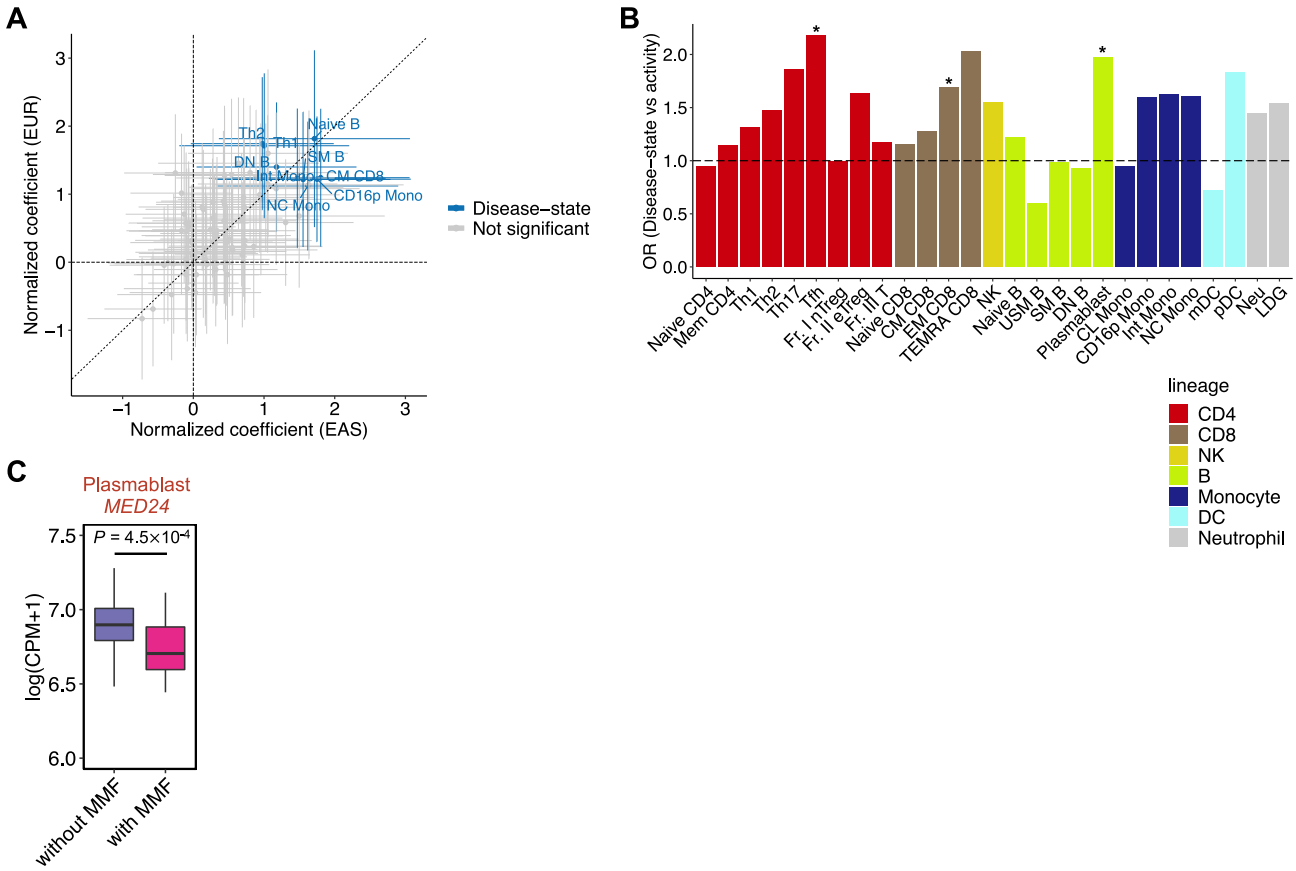


Figure S7. Risks variants for SLE are enriched around disease-state signatures, not disease-activity signatures, related to Figure 7

(A) Scatter plots comparing the normalized coefficients in S-LDSC for EAS and EUR SLE-GWAS. Each dot represents each signature (HC-SEGs, disease-state, and disease-activity signatures × 27 cell types = 81 signatures). The signatures that pass Bonferroni significance in meta-analysis are colored and annotated.

(B) A bar plot showing the direct comparison of SLE-GWAS prioritized genes enrichment between disease-state and disease-activity signature genes for each cell type using one-sided Fisher's exact test. Dashed horizontal line indicates odds ratio (OR) = 1. *nominal $p < 0.05$.

(C) A box plot showing the expression of *MED24* in plasmablasts between patients with and without MMF. p, p values in differential expression test. We analyzed the discovery dataset (n = 225) in this figure.